Detecting and quantifying change in extremes: a synthetic-data test of the Müller–Watson methods

Produced by Ken Caldeira with the help of Claude Code
https://github.com/Sustainable-Solutions-Lab/extreme-trend-detection

Draft, 3 November 2026

Summary

In brief. *Detection.* The parameter-constancy (Nyblom) test of Müller and Watson (2024) is a reliable detector of change in annual maxima: with one value per year its false-alarm rate stays at 4–6% for every parent distribution, record length and level of day-to-day persistence we tried. Its power depends strongly on the record length and on the shape of the daily distribution. A mean shift of half a daily standard deviation over the record is detected in 89% of 85-year records from a normal parent but in 24% from a skewed, precipitation-like (gamma) parent, and in 41% and 9% of 30-year records.

Quantification. Saying how large the change is, or what the current 1-in-100-year level is, is much harder. Even with no change and the correct model, the 99% annual-maximum level estimated from 85 years understates the true risk at least twofold in 27% of records (44% with 30 years). When extremes are changing, a GEV with a linear trend in location comes close to this floor (30% at 85 years), and no more flexible method, including the authors' Bayesian random-walk model, does better on average.

Using the wrong model. There is no single best model for the change. Whether to model it as a function of time or of the mean state depends on which actually drives it, and that is usually unknown. With 85 years, the wrong driving variable can raise the share of records in which the current 99% level misses the true risk by more than a factor of 2 by 10–20 percentage points: a mean-state model cannot see a change that does not move the mean, and a linear time trend cannot follow an accelerating mean state. When the mean accelerates, following its path matters more than getting right what drives the width, so the mean-state model loses little even when the width in fact changes on its own time course. Fitting a change that is not there costs about as much (11–16 points for the location + width and covariate models). Averaging the candidate models with Akaike weights is a robust way out: at 85 years the average is never more than 4 points worse than the best single model for the true process, while each single model is 11–30 points worse for some process. With 30 years, sampling noise dominates and no choice of model helps much.

Extended summary. We reimplemented the time-varying-extremes methods of Müller and Watson (2024) in Python, reproduced their published estimates and test statistics, and tested the methods on synthetic data for which the truth is known exactly. Daily series were drawn from four parent distributions (normal; gamma, precipitation-like; Student-t, heavy-tailed; beta, bounded)

with day-to-day correlation, and their mean, width or tail was changed over records of 30 to 150 years. With one value per year, all the tests we examined keep their nominal 5% false-alarm rate, except that the conventional location-trend likelihood-ratio test is over-sized (7–10%) for 30-year records. Power rises steeply with record length and is much lower for skewed or heavy-tailed parents, whose extremes are noisier relative to the daily variability. The Nyblom test is less powerful than a location-trend test against the change that test is built for, a linear shift in location, and much more powerful against changes in tail shape. Using the three or ten largest values per year adds power but, with strong day-to-day persistence, raises false alarms to 20–50% unless the extremes are declustered.

Detecting a change is not the same as measuring it. We scored a dozen estimators on a question that matters for risk assessment: if the estimated 99% annual-maximum level for the most recent year is taken at face value, how often is it really exceeded? An ideal estimator gives 1%. The dominant source of error is not the change but the difficulty of estimating a 1-in-100 level from a few decades of annual maxima: with no change and a correctly specified model, 27% of 85-year records understate the risk at least twofold and 18% overstate it at least twofold. Against this floor, a linear trend in location is the most reliable simple estimator (30% understated at 85 years). The authors' Bayesian random-walk model is competitive for 30- and 45-year records but worse for 85-year records (39%), it shrinks estimated changes toward zero, and its 90% credible intervals, while they cover the truth, are 1.3 to 3 times wider than the actual errors. With three values per year, 10–12% of its fits fail to converge under the authors' prior.

Which model is right depends on the physical process, and our tests show what happens when the model is wrong. We compared models of the same complexity that differ only in their driving variable: GEV location and scale linear in time, or linear in a smoothed annual mean of all daily values. When the distribution's width and tail follow an accelerating mean state, the time-trend models miss the current risk by more than a factor of 2 in 75–83% of 85-year records against 65% for the mean-state model. When the width changes while the mean stays constant, the mean-state model sees nothing and misses 69% against 55%. When the mean moves linearly in time the two coincide, and with 30 years every model misses 66–87% of records, so the choice of driving variable is secondary to record length. When the mean accelerates but the width and tail change linearly in time, the mean-state model, now with the wrong driver, still does slightly better (64% against 68%), because missing the curvature of the mean costs more than misplacing the width. The data rarely reveal the driver: AIC cannot tell a width change from a mean change. But an AIC-weighted average of the constant, time-trend and mean-state models comes within 4 points of the best single model for every true process at 85 years. The practical conclusion: these methods can establish *that* extremes are changing, and in which direction, but statements about the *size* of the current 1-in-100-year level should carry wide uncertainty, including the spread between physically plausible models of the change, and are best based on an average over those models rather than on one of them.

Contents

1 Questions **4**

2 Methods **4**

3 Detecting change **7**

4 Quantifying change **9**

5 Using the wrong model **22**

6 The Bayesian random-walk model **30**

7 Limitations **34**

8 Next steps **34**

A Cases and supplementary results **35**

1 Questions

Müller and Watson (2024) develop inference for extremes when parameters may vary over time: likelihood-ratio and Nyblom-type tests with small-sample critical values that depend on the estimated tail index, and a Bayesian time-varying GEV model with random walks in its parameters. Their simulations calibrate the size of the tests for data drawn exactly from the GEV model. They do not report power, do not test the methods on data whose extremes are only approximately GEV (the usual situation), say little about serial dependence, and do not check whether the Bayesian model recovers known parameter paths. For climate applications these are the questions that matter:

1. **Detection.** How often is a change of realistic size detected with record lengths typical of reanalyses and stations (30, 45, 85 and 150 years), and how often is a change "detected" when there is none?
2. **Quantification.** How well do the methods estimate the current return level, for example the level the annual maximum exceeds with probability 1% this year ($X_{0.99}$), and its change over the record?
3. **Using the wrong model.** A change can be modelled as a function of time (trends, random walks) or of the mean state (a covariate). Either could be physically right, and usually we do not know which. What is lost when the change follows one and we model it as a function of the other, and what can be done when we do not know?

2 Methods

2.1 Extremes model and tests

For each year $t = 1, \dots, T$ we keep the k largest daily values $X_t^{(1)} \geq \cdots \geq X_t^{(k)}$ (with $k = 1$, the annual maximum). By extreme-value theory these follow approximately the GEV_k distribution with location μ, scale σ and shape (tail index) ξ, with density

$$f(x_1, \dots, x_k) = \sigma^{-k} \exp\Big[-(1+\xi z_k)^{-1/\xi} \Big] \prod_{j=1}^{k} (1+\xi z_j)^{-1/\xi - 1}, \qquad z_j = \frac{x_j - \mu}{\sigma}.$$

The tests compared are:

- **Nyblom** (Müller and Watson, eq. 12; Nyblom, 1989): with scores S_t of the constant-parameter fit and average information V, $L_T = T^{-2} \sum_t \big(\sum_{l \geq t} S_l\big)' V^{-1} \big(\sum_{l \geq t} S_l\big)$. The test rejects when $L_T \exp(a_0 + a_1 \hat{\xi} + a_2 \hat{\xi}^2) > 1$, with coefficients calibrated by simulation so that the rejection rate is at most 5% for every true ξ on a grid (their Appendix B). We recalibrated the coefficients for each record length and k used here.
- **Location-trend LR**: GEV with $\mu_t = \mu_0 + \mu_1 t$, likelihood-ratio test of $\mu_1 = 0$ against χ^2_1. This is the standard tool in climate science (e.g. Coles, 2001).
- **Two halves**: separate GEV fits to the first and second halves of the record, LR test of equal parameters, critical values calibrated like the Nyblom test.
- **Mann–Kendall** rank test for a monotone trend in the annual maxima.

2.2 Estimators of the 99% level

Each estimator gives a path $\hat{X}_{0.99}(t)$, the level the annual maximum of year t exceeds with probability 1% (the 100-year return level in that year's climate). All are fitted by maximum likelihood to the same records, except the Bayesian model.

Constant GEV with constant parameters (ignores change).

Last 30 years constant GEV fitted to the most recent 30 years only, a common practical shortcut (equal to the constant fit for 30-year records).

Two halves constant GEV fitted to the second half of the record.

Location trend μ_t linear in time, σ and ξ constant.

Location + width trend μ_t and $\log \sigma_t$ linear in time, ξ constant.

All-parameter trend μ_t, $\log \sigma_t$ and ξ_t linear in time.

Mean-state covariate μ_t and $\log \sigma_t$ linear in c_t, ξ constant, where c_t is a smoothed annual mean of *all* daily values (local-linear smoother, Gaussian kernel, bandwidth 10 years, fixed in advance). This is the usual approach in event attribution, where the covariate is a smoothed (global or local) mean temperature (Philip et al., 2020). A variant with all three parameters linear in the raw annual mean is unstable (the tail index follows the year-to-year noise of the covariate) and is shown only for reference.

Covariate (true mean) as above but with the true mean path as covariate: an oracle that shows the cost of estimating the covariate.

Data-chosen (AIC) for when the form of the change is unknown: of the constant, location-trend, location + width trend and mean-state covariate models, the one with the lowest AIC.

AIC-weighted average the average of the same four models' $X_{0.99}$ paths, with Akaike weights $w_i \propto \exp(-\Delta\mathrm{AIC}_i/2)$.

Bayesian the authors' four-random-walk model (their `GEV_4RW_Model.stan`, unchanged likelihood and priors), posterior median of $X_{0.99}(t)$. The parameters follow random walks in a logistic transform of $\xi_t \in (-0.9, 2.1)$, in $\ln \alpha_t$, $\ln s_t$ and m_t, with $\sigma_t = s_t \alpha_t^{\xi_t}$ and $\mu_t = m_t + (s_t/\xi_t)(\alpha_t^{\xi_t} - 1)$. The innovation standard deviations are $\gamma = \mathrm{sg} \cdot g/\sqrt{T}$ with $g \sim \mathrm{Exp}(1)$ and scale factors $\mathrm{sg} = 1, 0.5, 0.5$ for ξ, α, s (and the scale level for m). We use 4 chains of 1,000 warm-up and 1,000 sampling iterations (`adapt_delta` 0.95); the authors used 5,000 and 10,000 for single published fits. We also report the posterior mean, and a run with a tighter prior ($\mathrm{sg} = 0.25$ for all three).

2.3 Synthetic data

A daily standard Gaussian series Z_d with AR(1) correlation ϕ (0.5 unless stated; 0 and 0.8 also run) is mapped through the parent's quantile function, $X_d = m_t + s_t\, \varepsilon(Z_d; \mathrm{tail}_t)$, with m_t, s_t and the tail parameter constant within a year. The parents are standardized so that the baseline daily standard deviation is 1; all levels are in these units. For a daily temperature with a standard deviation of 3–4 K, a change of 0.5 is 1.5–2 K.

Parent	ε	Tail parameter (baseline)	Tail of the extremes
Normal	Z	—	light, $\xi \to 0$ slowly
Gamma	standardized gamma(a)	shape $a = 2$	exponential-type, skewed
Student-t	standardized t_ν	$\nu = 6$	heavy, $\xi = 1/\nu$
Beta	$3(2B - 1)$, $B \sim \text{beta}(\alpha, \beta)$	$\alpha = \beta = 4$	bounded, $\xi = -1/\beta$

Because the map from Z to X is increasing within a year, the annual maximum of X is the map applied to the annual maximum of Z, so the true $X_{0.99}(t)$ is known exactly: in closed form for independent days, and from 500,000 simulated years of Z for correlated days (error about 0.005 at $p = 0.99$).

The changes (Appendix, Table 4) move one feature at a time over the record, along a linear, mid-record step or accelerating (exponential) path: the mean, the width ($\log s$), the tail parameter (log of a, ν or β), or, for the beta, the position of the mass within fixed bounds. A second set, the *covariate-model cases*, is built so that a mean-state model is correct:

- *Temperature-like* (normal, t, beta): the mean rises by 0.5 (beta 0.25) along an accelerating path, the width follows it ($\log s_t = 0.25\, m_t$), and for the t and beta so does the tail ($\log \nu_t$ or $\log \beta_t = -0.2\, m_t$, a heavier-tailed t and a more tightly bounded beta as it warms).
- *Precipitation-like* (gamma): the mean is scaled by $\lambda_t = e^{g_t}$ with a 22% rise over the record, so every quantile scales with the mean (a Clausius–Clapeyron-type change); in a variant the gamma shape also falls as it gets wetter (heavier tail).

A third set, the *time-driven* counterparts of the accelerating temperature-like cases, has the same mean path and the same first- and last-year distributions, but the width and tail change linearly in time between those states instead of following the mean (Section 5).

2.4 Scoring

For detection we report the rejection rate at the nominal 5% level with no change (size) and with change (power). For quantification, the main score is the *true exceedance probability* of each record's estimated level in the last year,

$$\pi = \Pr\left(M_T > \hat{X}_{0.99}(T)\right),$$

computed from the exact distribution of the last year's annual maximum M_T. An ideal estimator has $\pi = 1\%$. We report its median over records and the shares of records with $\pi > 2\%$ (risk understated at least twofold) and $\pi < 0.5\%$ (overstated at least twofold). This score is comparable across parents and penalizes errors by their consequence for risk rather than by their size in data units. We also report the error of the estimated change $\hat{X}_{0.99}(T) - \hat{X}_{0.99}(1)$ and, for the Bayesian model, the coverage and width of its 90% credible intervals.

2.5 Monte Carlo design

Detection results ("Stage B" in the repository) use 10,000 simulated records with no change and 2,000 per change. Quantification results ("Stage C") use 100 records per case, fitted by all estimators. The same daily noise is used across cases, record lengths and estimators (common random numbers), so comparisons between estimators are paired. A bootstrap over records gives standard

errors of 3–4 percentage points on the pooled shares and about 3 points on differences between estimators.

3 Detecting change

3.1 False alarms

With one value per year, all four tests hold their nominal size within Monte Carlo error at every record length and persistence level (Figure 1), except the location-trend LR test for short records (7–10% at $T = 30$), where the χ^2 approximation is poor. Using several values per year from a persistent daily series breaks the independence the GEV_k model assumes: with the three largest values per year, the Nyblom test rejects 7–9% at $\phi = 0.5$ and 21–23% at $\phi = 0.8$, and with ten values 46–51%. Requiring the values to be at least three days apart (declustering) brings the last case down to 9–12%. Mann–Kendall uses only the annual maxima and is unaffected.

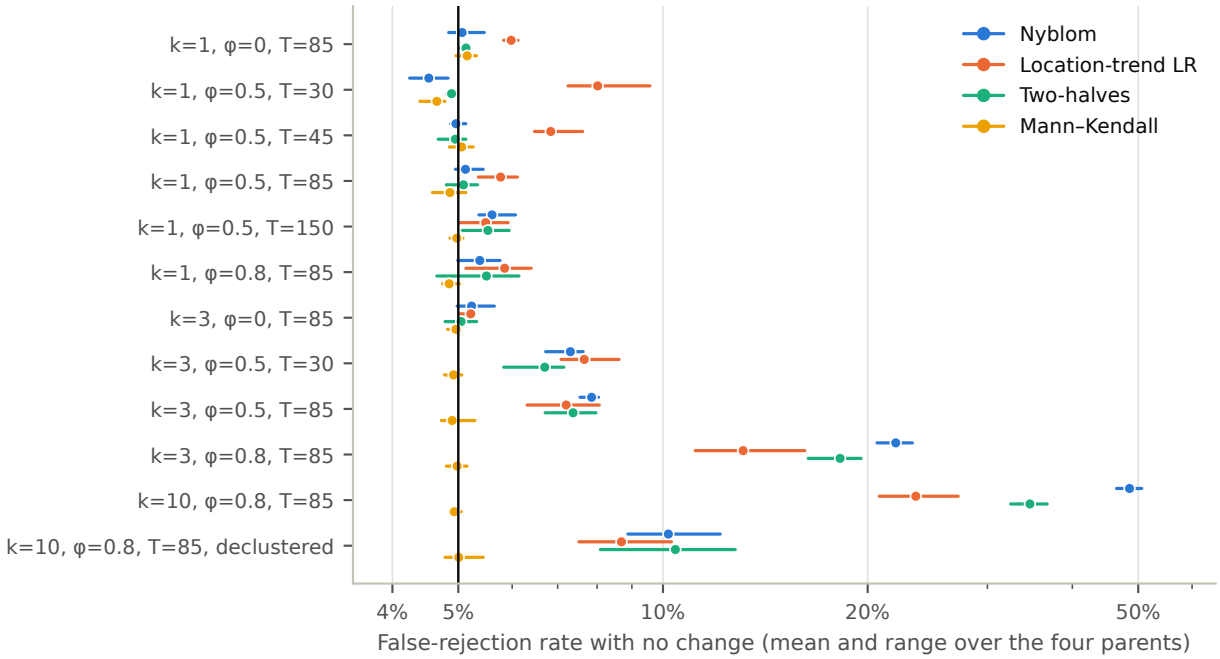


Figure 1: False-rejection rate of the four tests with no change, by design (k values per year, day-to-day correlation ϕ, record length T). Dots: mean over the four parents; lines: range. Vertical line: nominal 5%.

3.2 Power

Power rises steeply with record length (Figure 2, Table 1). For a normal parent, a mean shift of 0.5 daily standard deviations over the record is detected by the Nyblom test in 41%, 60%, 89% and 99% of records of 30, 45, 85 and 150 years. The same shift is much harder to detect for skewed or heavy-tailed parents (gamma: 9–41%; t: 11–52%), whose annual maxima are more variable relative to the daily standard deviation. The location-trend LR test is more powerful against the change

it is designed for, a linear shift in location (63% at 30 years and 96% at 85 years for the normal mean shift, though part of the edge at 30 years comes from its over-sizing). The Nyblom test is much more powerful against a change in tail shape (t parent: 69% vs 36% at 85 years). The two tests are complementary rather than competing.

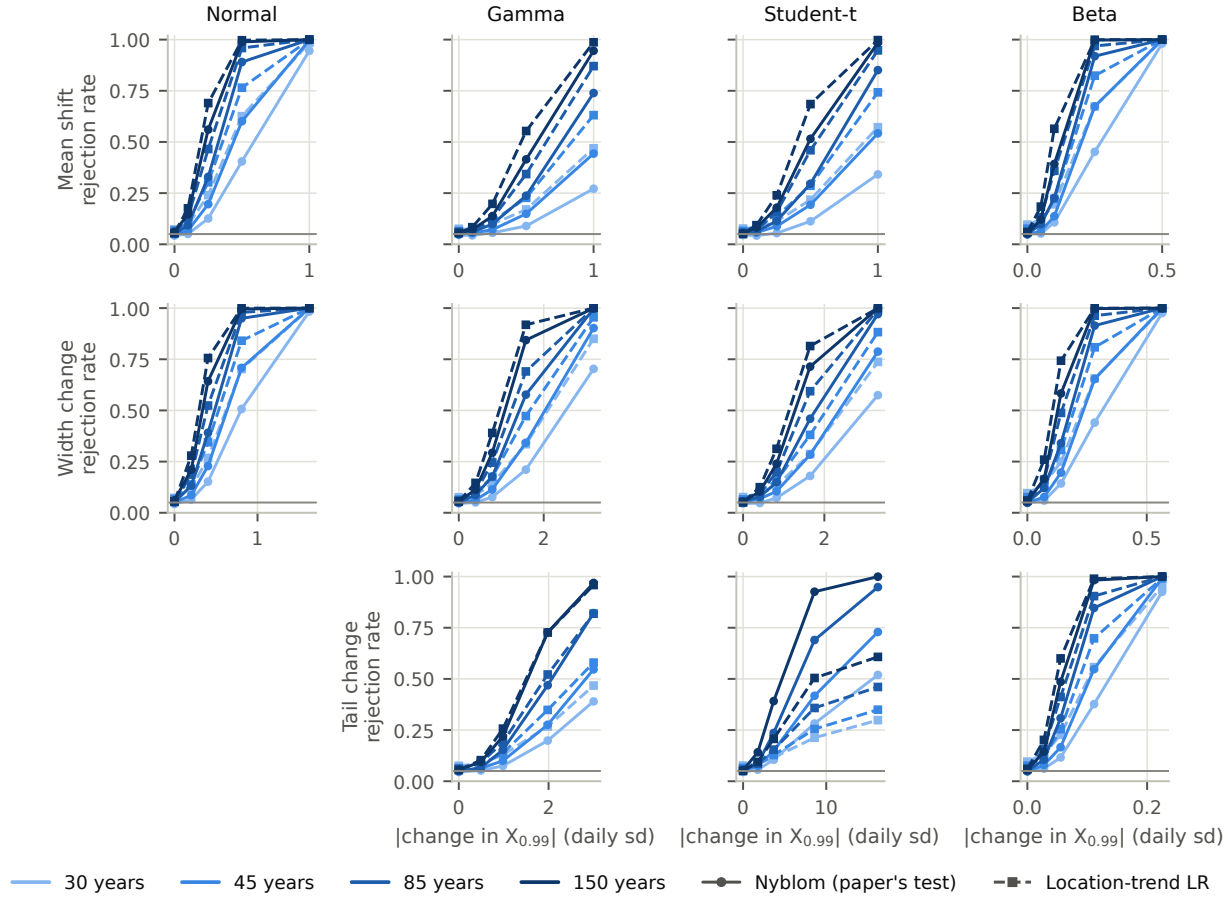


Figure 2: Rejection rate (nominal 5% level) against the size of the true change in $X_{0.99}$ (computed for independent days; the correlated-day value differs by under 1%), for linear changes in mean, width and tail, one value per year, $\phi = 0.5$. Colour: record length. Solid: Nyblom; dashed: location-trend LR.

			$T = 30$		$T = 45$		$T = 85$		$T = 150$	
Parent	Change	$\Delta X_{0.99}$	Nyb.	LR	Nyb.	LR	Nyb.	LR	Nyb.	LR
Normal	mean 0.5	+0.50	0.41	0.63	0.60	0.77	0.89	0.96	0.99	1.00
Normal	width 0.2	+0.81	0.51	0.70	0.71	0.84	0.95	0.98	1.00	1.00
Gamma	mean 0.5	+0.50	0.09	0.17	0.15	0.23	0.24	0.34	0.41	0.55
Gamma	width 0.2	+1.58	0.21	0.34	0.34	0.47	0.58	0.69	0.84	0.92
Gamma	tail 1	-1.98	0.20	0.27	0.28	0.35	0.47	0.52	0.73	0.73
Student-t	mean 0.5	+0.50	0.11	0.22	0.19	0.29	0.30	0.46	0.52	0.69
Student-t	tail 1	-8.59	0.28	0.21	0.42	0.26	0.69	0.36	0.93	0.50
Beta	mean 0.25	+0.25	0.45	0.67	0.67	0.82	0.92	0.97	1.00	1.00
Beta	width 0.1	+0.28	0.44	0.66	0.65	0.81	0.91	0.96	1.00	1.00
Beta	tail 0.2	-0.11	0.38	0.56	0.55	0.70	0.85	0.90	0.98	0.99
Beta	alpha 0.5	+0.07	0.31	0.45	0.49	0.59	0.80	0.82	0.98	0.97
All (mean)	no change	0	0.045	0.080	0.050	0.068	0.051	0.058	0.056	0.055

Table 1: Power (nominal 5% level) of the Nyblom (Nyb.) and location-trend LR tests for selected linear changes, one value per year, $\phi = 0.5$. $\Delta X_{0.99}$: true change in the 99% level over the record (baseline daily sd). Last row: size with no change, mean over parents.

4 Quantifying change

4.1 What the estimates look like

Figure 3 shows single 85-year records with the estimated and true paths of $X_{0.99}$. The records were chosen by a fixed rule, as the record of median difficulty (ranking records by the average over the estimators of $|\log(\pi/1\%)|$), so they are typical rather than best or worst cases. Figure 4 shows the spread of the estimated paths over all 100 records for three cases, and Figure 5 the effect of record length.

Three features stand out. First, the estimates scatter widely about the truth: for a normal parent with 85 years and a mean shift of 0.5, the 5–95% range of the error of the last year's $X_{0.99}$ spans 0.7 daily standard deviations for the location trend and about 1.0 for the more flexible estimators, larger than the change being estimated. Second, a model with the wrong form for the change is biased in a predictable way: the constant fit is flat, and the location trend, a straight line, misses the late acceleration in the covariate-model cases. Third, the Bayesian posterior median follows the truth on average but with paths that are too flat toward the end of the record, and with 30 years its credible band is very wide upward.

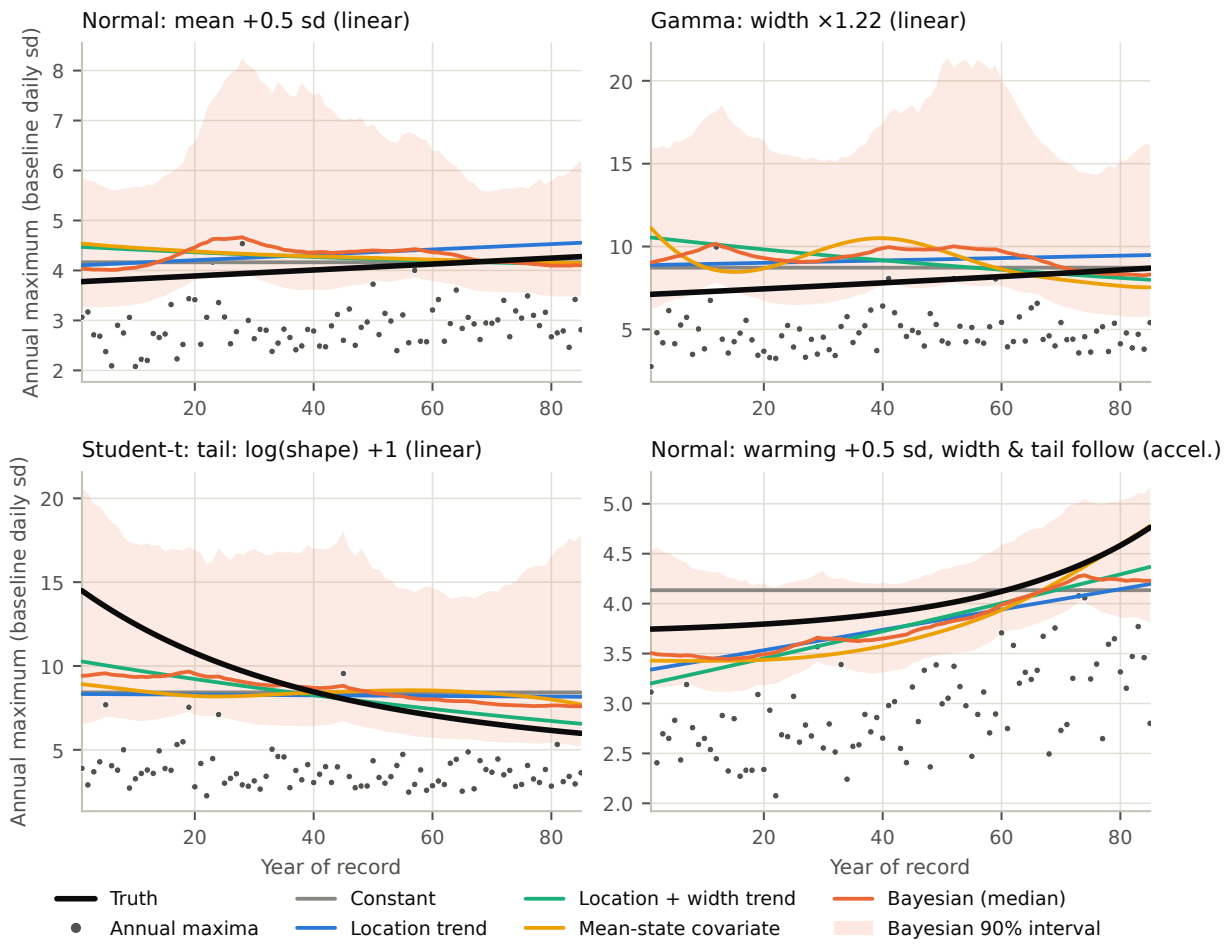


Figure 3: Four typical 85-year records (points: annual maxima) with the true 99% level (black) and five estimates. Shaded: Bayesian 90% credible interval. Units: baseline daily standard deviation.

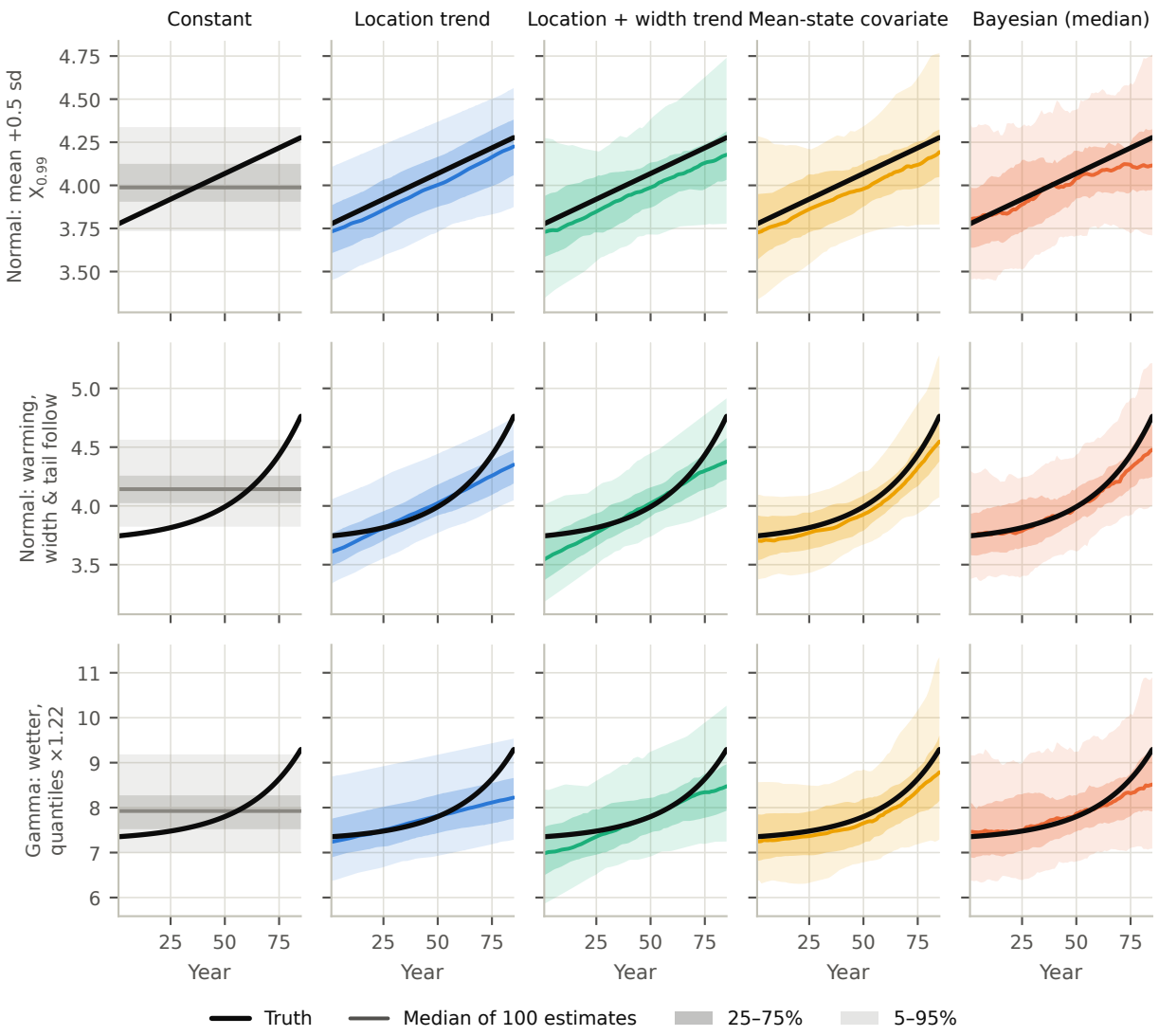


Figure 4: Spread of the estimated 99% level over 100 records of 85 years (median, 25–75% and 5–95% of the estimates in each year) for a linear mean shift (top) and two covariate-model cases (middle, bottom), with the truth in black.

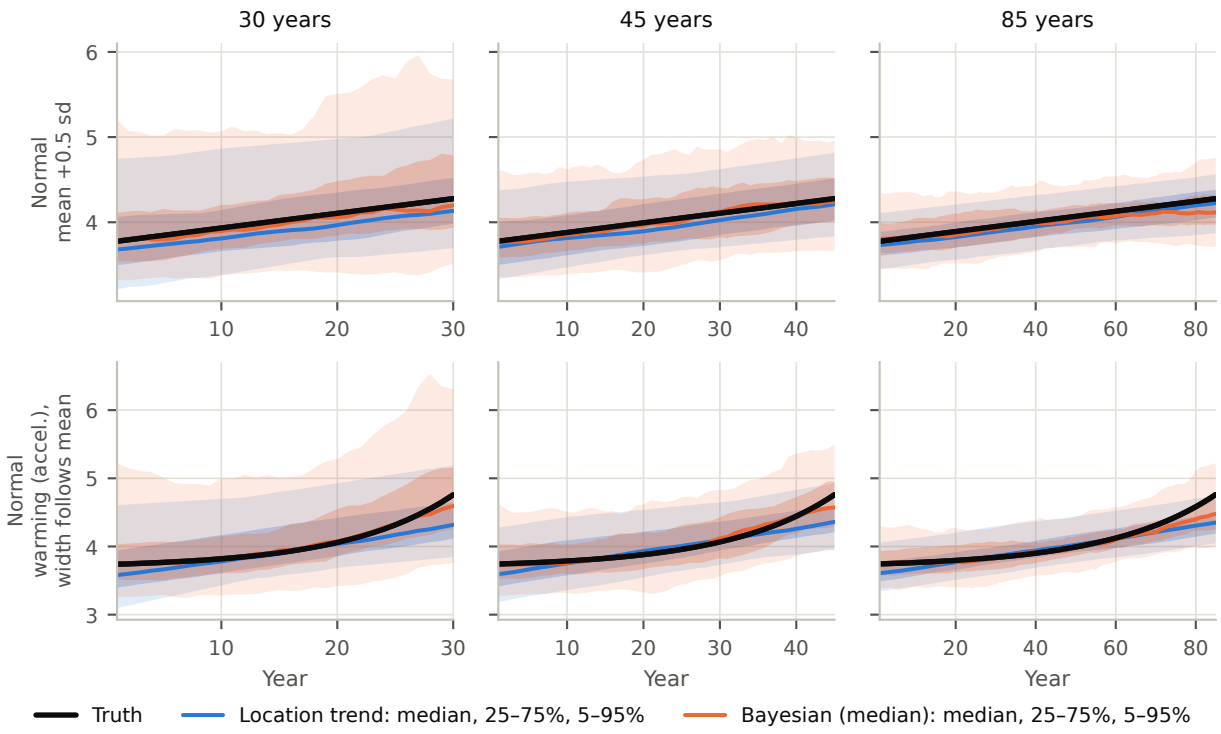


Figure 5: Effect of record length: the location-trend (blue) and Bayesian (orange) estimates for 30, 45 and 85 years. The total change over the record is the same in each column, so the change per year is larger for shorter records.

4.2 Estimated paths, case by case

Figures 6–9 repeat Figure 5 for every standard case, one figure per parent, comparing the location trend (blue) with the Bayesian posterior median (orange). Four patterns recur.

- **Mean and width changes are tracked on average by both estimators**, in all four parents. What changes with record length is the spread: the 5–95% band at 85 years is 37–50% as wide as at 30 years.
- **A step change is smoothed into a ramp.** Neither estimator locates the step; the location trend replaces it by a straight line, and the Bayesian median by a gentler rise.
- **Short records from skewed or heavy-tailed parents give the Bayesian model a long upper tail.** At 30 years the upper edge of its 5–95% band for the gamma and t parents lies far above the truth (up to about 2.5 times the true level for the t), because the posterior of the tail index is wide and right-skewed when there are few maxima.
- **Tail changes are where the two estimators differ most.** The location trend cannot represent a change in tail: it captures about a third of the fall in the 99% level for the gamma parent and almost none (5–7%) for the t. The Bayesian median captures more, 51–64% for the gamma and 37–50% for the t, increasing with record length. In the bounded beta parent the location trend does the opposite and overshoots the tail and position changes (by a factor

of about 1.9 for the tail change), presumably because a straight-line location with fixed scale and shape is the only way it can absorb a change in the shape of the maxima near the bound.

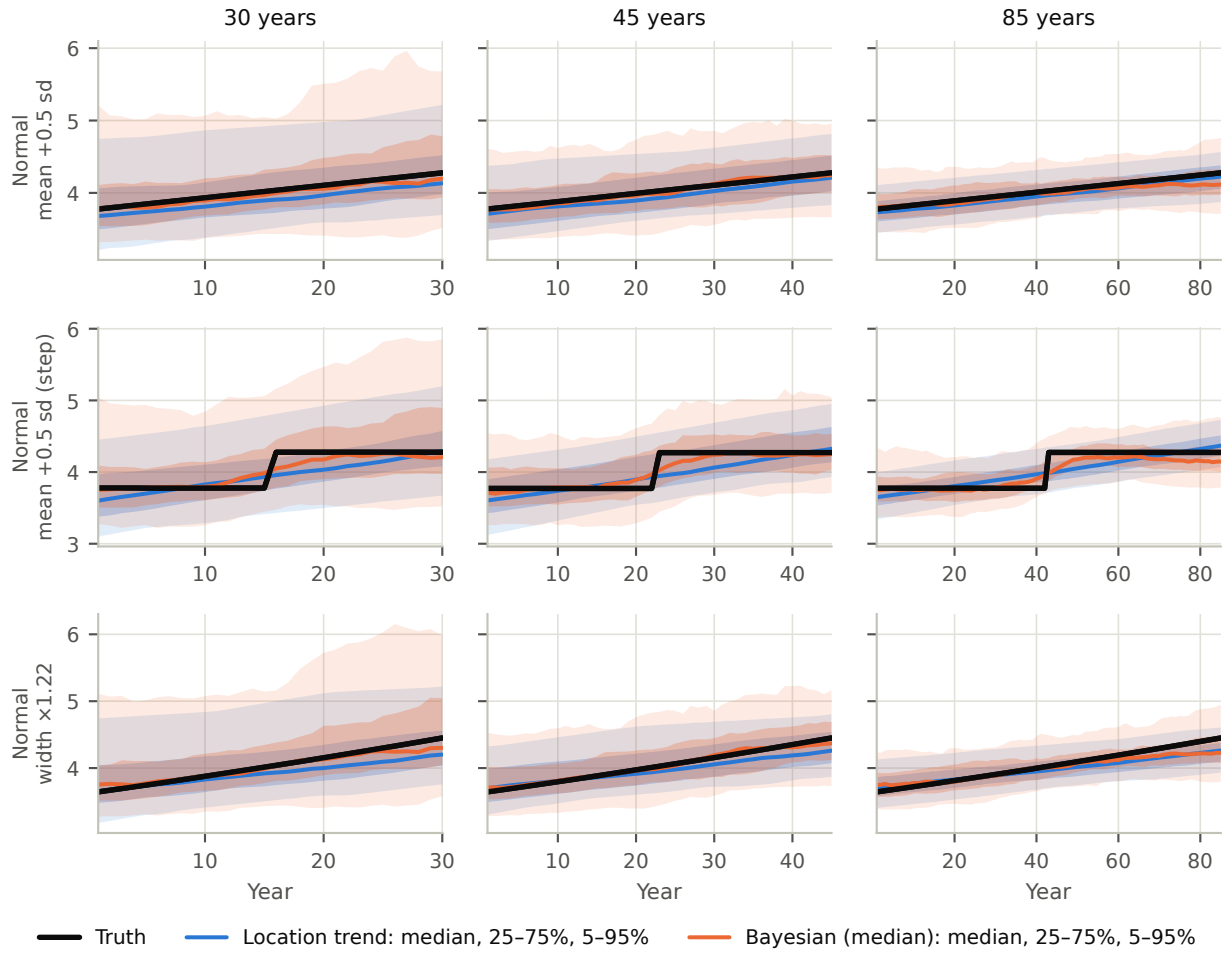


Figure 6: Normal parent, all standard changes. Median (line), 25–75% (dark band) and 5–95% (light band) of the estimated 99% level in each year over 100 records, with the truth in black. Columns: record length; the total change over the record is the same in each column. Units: baseline daily sd.

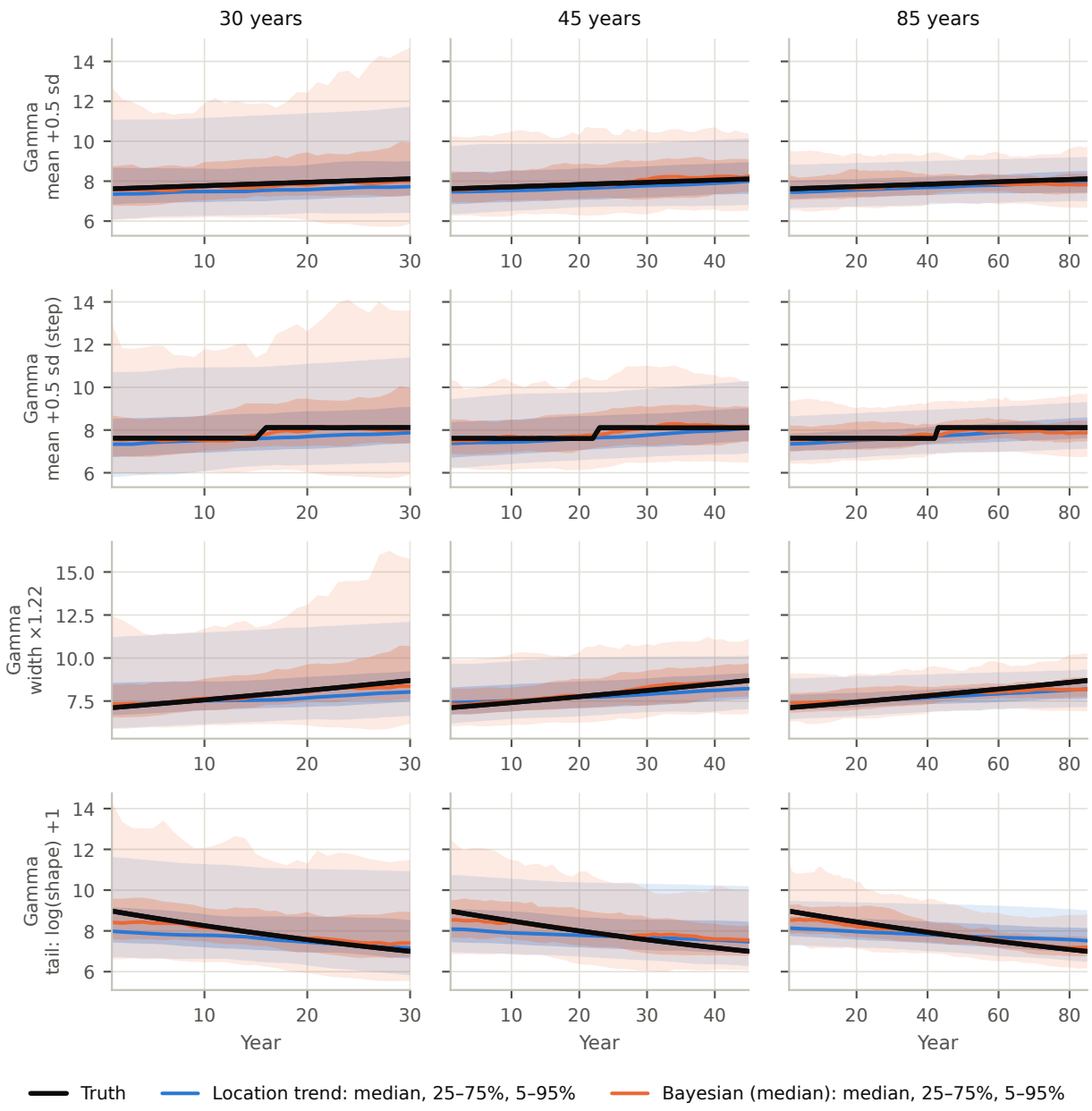


Figure 7: Gamma (precipitation-like) parent, all standard changes. Median (line), 25–75% (dark band) and 5–95% (light band) of the estimated 99% level in each year over 100 records, with the truth in black. Columns: record length; the total change over the record is the same in each column. Units: baseline daily sd.

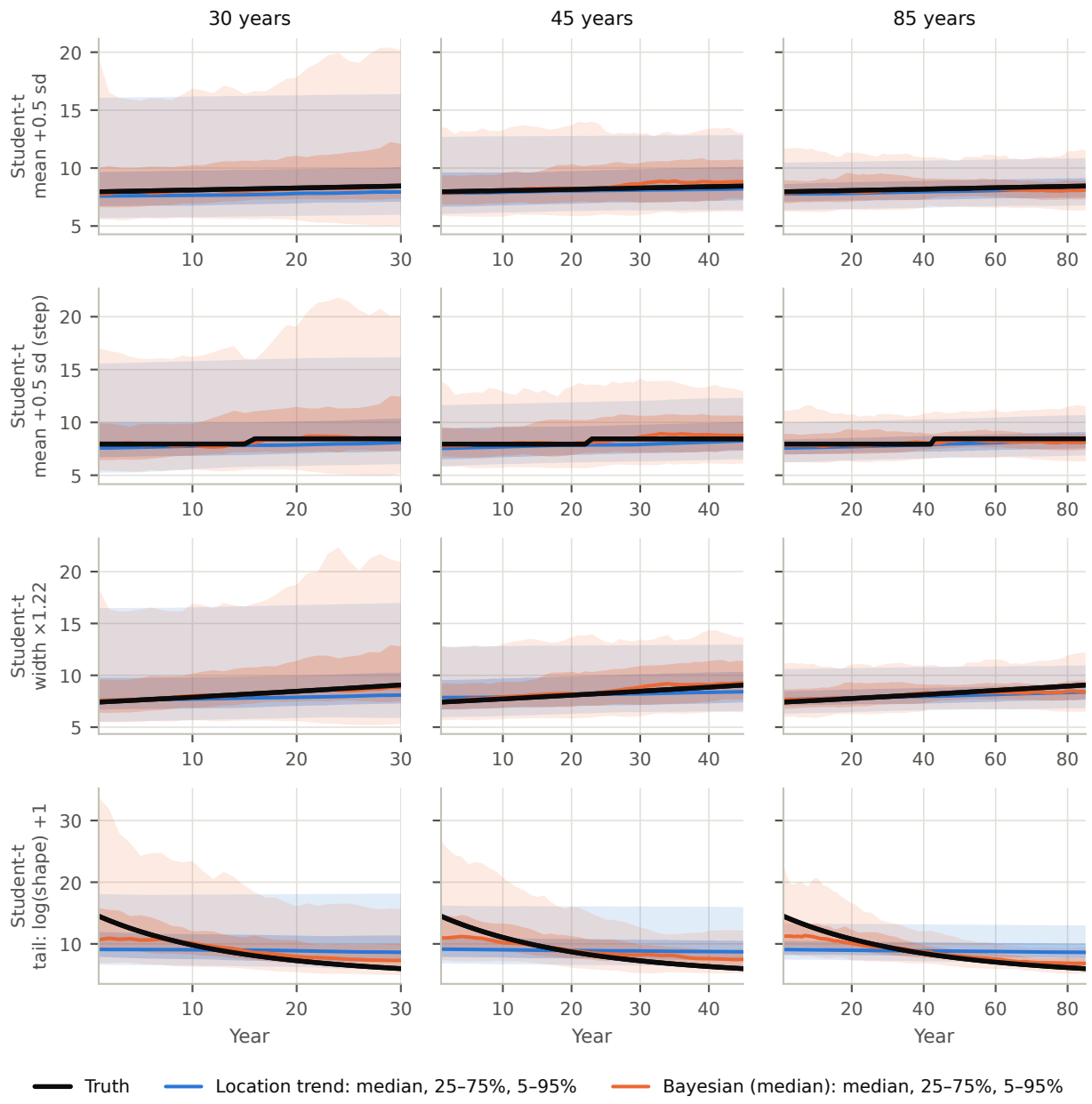


Figure 8: Student-t (heavy-tailed) parent, all standard changes. Median (line), 25–75% (dark band) and 5–95% (light band) of the estimated 99% level in each year over 100 records, with the truth in black. Columns: record length; the total change over the record is the same in each column. Units: baseline daily sd.

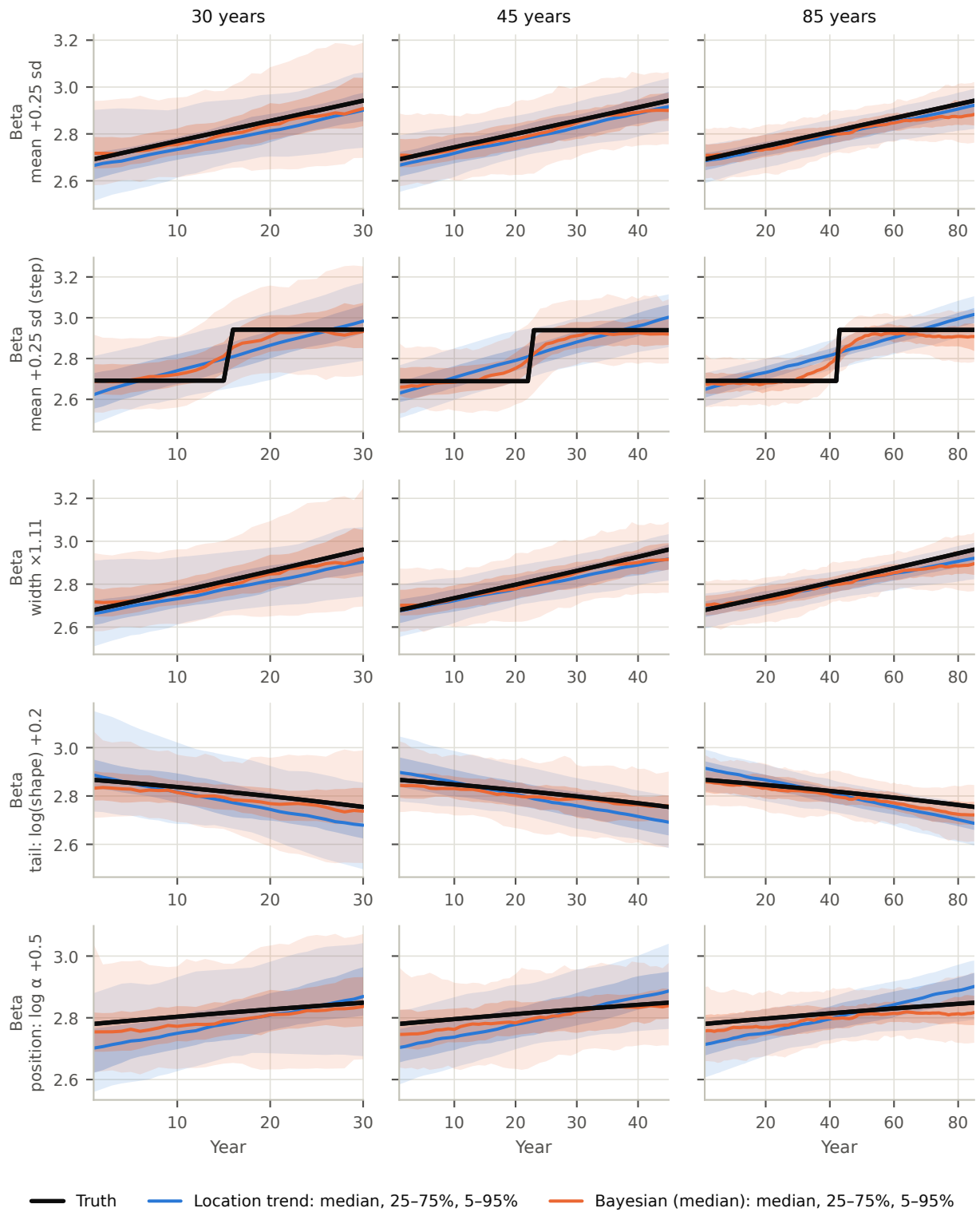


Figure 9: Beta (bounded) parent, all standard changes. Median (line), 25–75% (dark band) and 5–95% (light band) of the estimated 99% level in each year over 100 records, with the truth in black. Columns: record length; the total change over the record is the same in each column. Units: baseline daily sd.

4.3 The noise floor

Much of the error has nothing to do with change. With no change and the correctly specified constant model, the estimated $X_{0.99}$ understates the risk at least twofold in 27% of 85-year records and overstates it at least twofold in 18%; for 30-year records the shares are 44% and 27%, and for 45-year records 38% and 30%. The median exceedance probability is 1.3% rather than 1%: the sampling distribution of an estimated high quantile is skewed, so the typical estimate is a little low. This floor is the yardstick for the results below. An estimator that allows for change should not be expected to beat it, and it pays for its flexibility with extra variance when the change is small.

4.4 Accuracy of the current 99% level

Figures 10 and 11 and Table 2 pool the results over the standard cases (20 cases including the four with no change) and the covariate-model cases (7 cases).

- **The location trend is the most reliable simple choice** for the standard cases: at 85 years it understates the risk twofold in 30% of records, close to the 27% floor. Ignoring change (constant, 46%) or using only the last 30 years (49%) is much worse.
- **More flexible time trends do not help on average.** Adding a width trend gives 33% and trends in all three parameters 42%; the gain where the width or tail really changes is outweighed by the extra variance elsewhere.
- **The Bayesian model is competitive for short records but not for long ones**: 36% at 30 years (against 40% for the location trend), 34% at 45 years, but 39% at 85 years, which is 10 ± 3 points worse than the location trend. Its posterior mean is a poor point estimate: the posterior of a high quantile is right-skewed, so the mean overstates the level (median exceedance 0.5% at 85 years, and below 0.01% at 30 years, where the posterior of a high quantile has a very long right tail).
- **Three values per year help modestly** (location trend 28% at 85 years, Bayesian 31%), when declustered. Figure 12 shows the bands narrowing a little, and the Bayesian median following the t tail change more closely; it also shows the unconverged Bayesian fits for the beta parent (Section 6) as a wide upper band early in the 85-year record.
- **Record length matters more than the method.** Every estimator is worse at 30 years than at 85, and the spread between the best and worst sensible methods (about 10 points) is smaller than the floor itself.

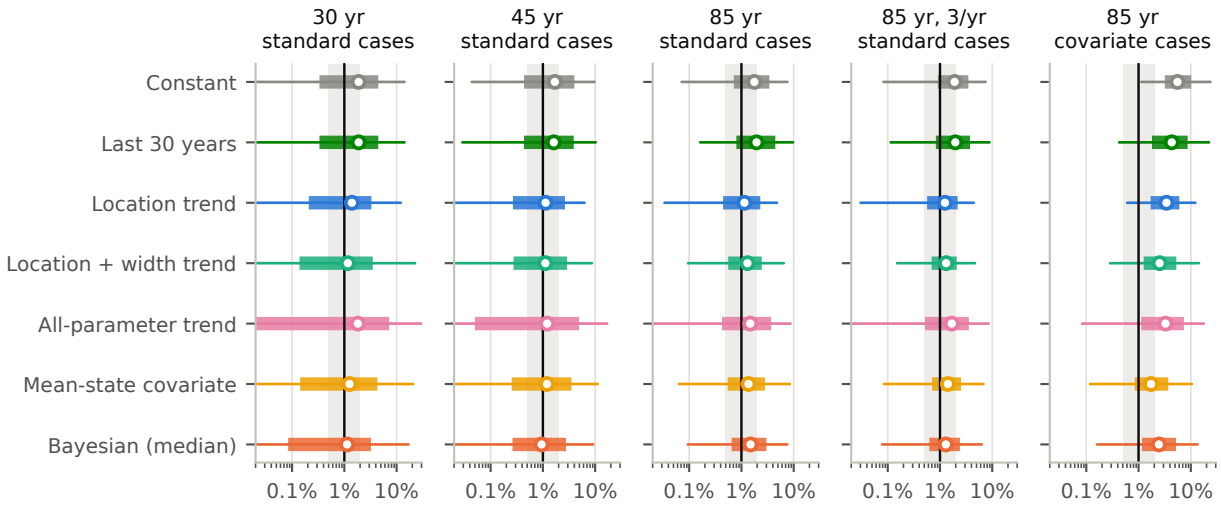


Figure 10: Distribution over records of the true probability that the estimated 99% level is exceeded in the last year (circle: median; bar: 25–75%; line: 5–95%). Shaded band: within a factor 2 of the ideal 1%.

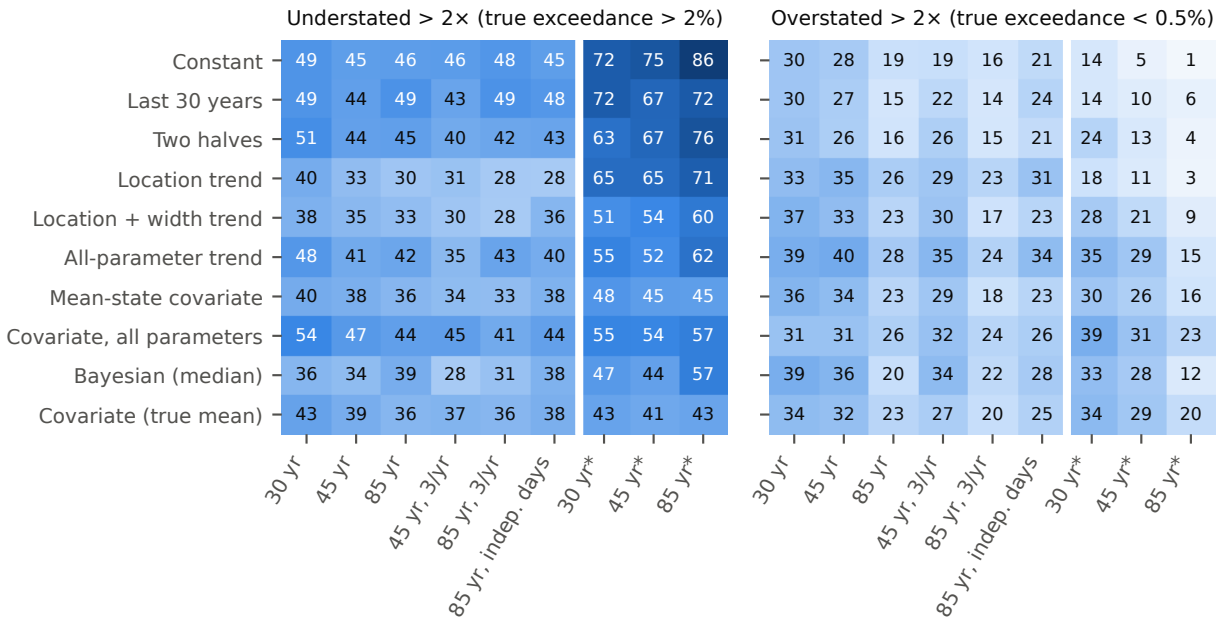


Figure 11: Share of records in which the estimated 99% level understates (left) or overstates (right) the true risk at least twofold, pooled over cases, by design. Columns marked * are the covariate-model cases.

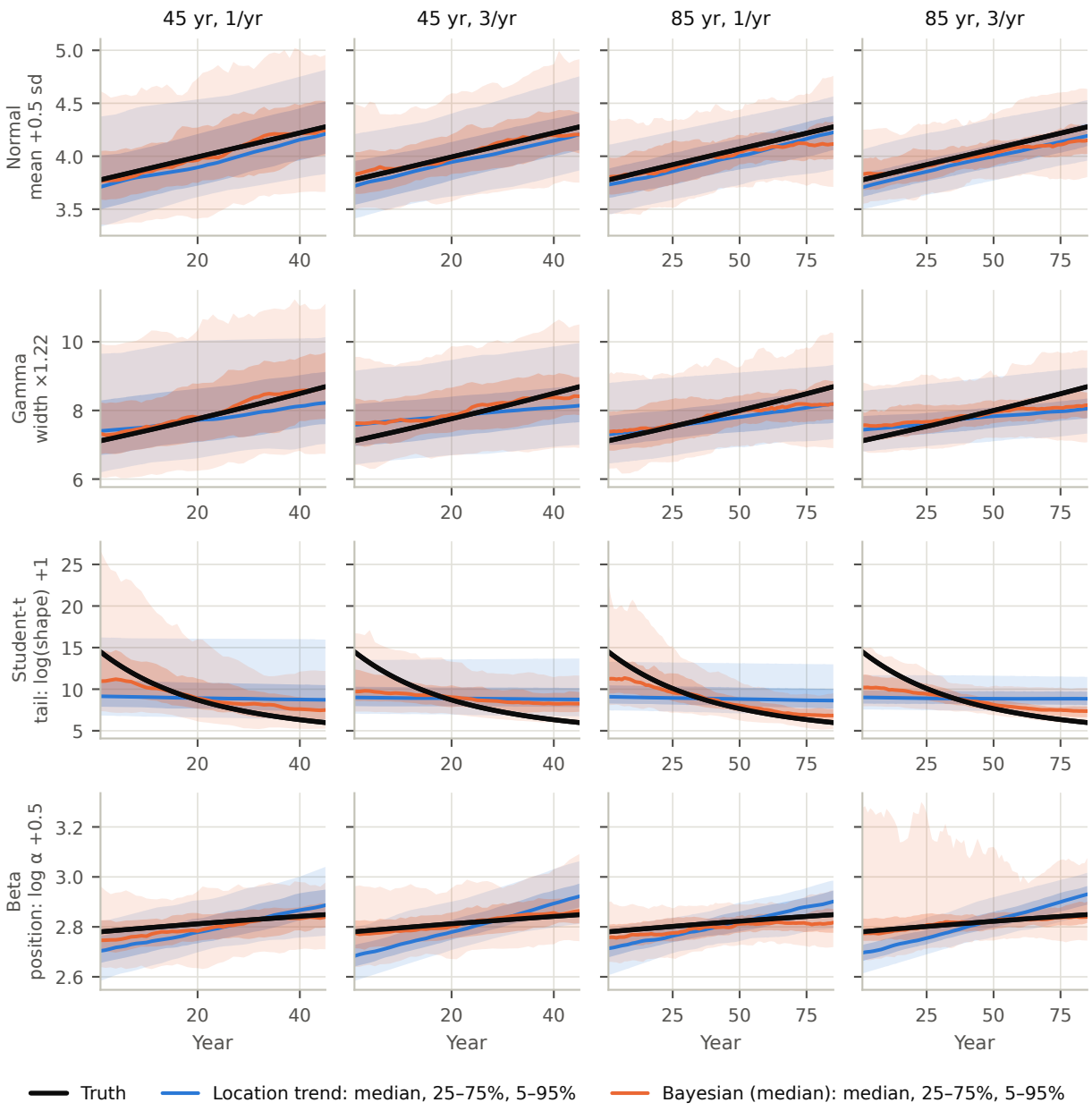


Figure 12: One against three values per year (declustered, at least three days apart), at 45 and 85 years, for four cases. Median (line), 25–75% (dark band) and 5–95% (light band) of the estimated 99% level in each year over 100 records, with the truth in black. Units: baseline daily sd.

	30 yr			45 yr			85 yr			85 yr, 3/yr		
Estimator	med.	>2%	<0.5%	med.	>2%	<0.5%	med.	>2%	<0.5%	med.	>2%	<0.5%
Standard cases												
Constant	1.88	49	30	1.69	45	28	1.77	46	19	1.91	48	16
Last 30 years	1.88	49	30	1.62	44	27	1.94	49	15	1.96	49	14
Two halves	2.07	51	31	1.67	44	26	1.75	45	16	1.68	42	15
Location trend	1.39	40	33	1.13	33	35	1.14	30	26	1.24	28	23
Location + width trend	1.17	38	37	1.12	35	33	1.30	33	23	1.31	28	17
All-parameter trend	1.81	48	39	1.21	41	40	1.47	42	28	1.68	43	24
Mean-state covariate	1.26	40	36	1.20	38	34	1.37	36	23	1.41	33	18
Covariate, all parameters	2.57	54	31	1.75	47	31	1.64	44	26	1.51	41	24
Data-chosen (AIC)	1.48	42	33	1.27	38	33	1.31	34	21	1.38	30	16
AIC-weighted average	1.47	41	34	1.17	36	33	1.30	30	20	1.40	28	15
Bayesian (median)	1.13	36	39	0.94	34	36	1.50	39	20	1.30	31	22
Bayesian (mean)	0.00	5	87	0.01	10	78	0.48	20	51	0.79	22	38
Covariate (true mean)	1.56	43	34	1.31	39	32	1.43	36	23	1.48	36	20
Covariate, all par. (true mean)	1.63	45	35	1.38	41	33	1.48	40	25	1.57	39	22
Covariate-model cases												
Constant	5.49	72	14	5.34	75	5	5.56	86	1			
Last 30 years	5.49	72	14	4.14	67	10	4.32	72	6			
Two halves	4.17	63	24	4.15	67	13	4.42	76	4			
Location trend	3.58	65	18	3.19	65	11	3.42	71	3			
Location + width trend	2.20	51	28	2.32	54	21	2.53	60	9			
All-parameter trend	2.94	55	35	2.25	52	29	3.28	62	15			
Mean-state covariate	1.85	48	30	1.65	45	26	1.73	45	16			
Covariate, all parameters	3.72	55	39	2.65	54	31	2.83	57	23			
Data-chosen (AIC)	2.68	57	25	2.18	53	21	2.04	50	12			
AIC-weighted average	2.45	56	26	2.20	52	21	2.09	52	12			
Bayesian (median)	1.67	47	33	1.67	44	28	2.46	57	12			
Bayesian (mean)	0.00	5	89	0.01	12	77	0.69	28	45			
Covariate (true mean)	1.40	43	34	1.39	41	29	1.61	43	20			
Covariate, all par. (true mean)	2.23	52	36	1.46	46	39	2.01	50	29			

Table 2: Accuracy of the estimated 99% level in the last year: median true exceedance probability (%, ideal 1) and shares of records (%) with the risk understated (> 2%) or overstated (< 0.5%) at least twofold.

4.5 Estimating the size of the change

Figure 13 shows the estimated change in $X_{0.99}$ from the first year to the last for every case. The location trend is precise but estimates only the part of the change that is a shift in location: it captures between a third and four-fifths of a width change (depending on the parent) and a third or less of a tail change in the gamma and t parents. The location + width trend and the covariate model are roughly unbiased where their form is right, but their 10–90% ranges are as wide as the change itself. The Bayesian model systematically shrinks the change toward zero, most visibly for the large tail change in the t parent and for the precipitation-like cases. For a normal parent with 85 years and a true change of 0.5, the location trend's 10–90% range is 0.33–0.65. The location + width trend and the covariate model span about 0 to 0.9, so even the sign is wrong in about one record in ten. The Bayesian median change is 0.32, against the true 0.5.

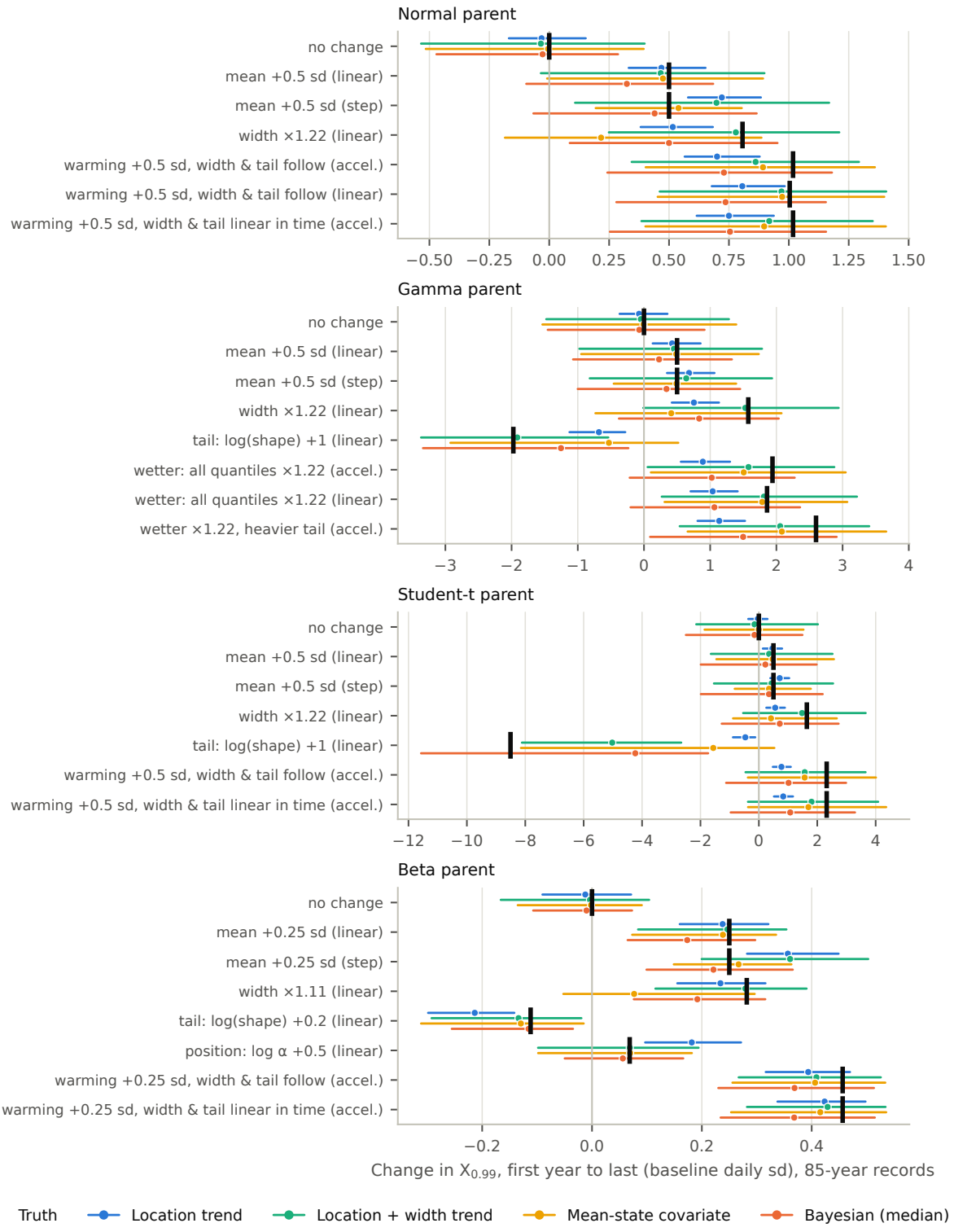


Figure 13: Estimated change in the 99% level over 85-year records (median and 10–90% over records) against the truth (black bar), for every case.

5 Using the wrong model

A change in extremes can be modelled as a function of time (a constant, a linear trend in location, linear trends in location and width, or a random walk) or as a function of the mean state, here a smoothed annual mean of all daily values. Physically, either could be right. The width of the temperature distribution may change because the climate has warmed, a function of the mean state, or for reasons with their own time course, such as changes in circulation, land use or aerosols, a function of time. Usually we do not know which. There is therefore no single right model. The useful questions are what is lost when the model's driving variable is the wrong one, and what to do when it is unknown.

When the two kinds of model differ. They coincide when the mean state moves linearly in time, and they differ when it does not (a step, an acceleration) or when the distribution changes without any change in the mean. We classified the cases by their true process (Table 3) and compare two models with the same number of parameters that differ only in their driving variable: the location + width trend (μ and $\log\sigma$ linear in time) and the mean-state covariate model (μ and $\log\sigma$ linear in the smoothed annual mean). To complete the comparison we added time-driven counterparts of the accelerating warming cases: the same mean path and the same first- and last-year distributions, but with the width and tail changing linearly in time instead of following the mean. In each pair the two processes differ only in how the distribution gets from its first state to its last (Figure 15). At the coupling strength we use, the width in mid-record differs by about 4% between the two.

Score. The tail changes here all lower the 99% level and the mean changes all raise it, so the one-sided score of Section 4 favours models that err in the direction of the change: a constant fit, for instance, never understates the risk after a tail change. For this comparison we therefore count the records in which the estimated level misses the true risk by more than a factor of 2 *in either direction* (true exceedance probability above 2% or below 0.5%). With no change and the correct model, this happens in 44% of 85-year records and 71% of 30-year records.

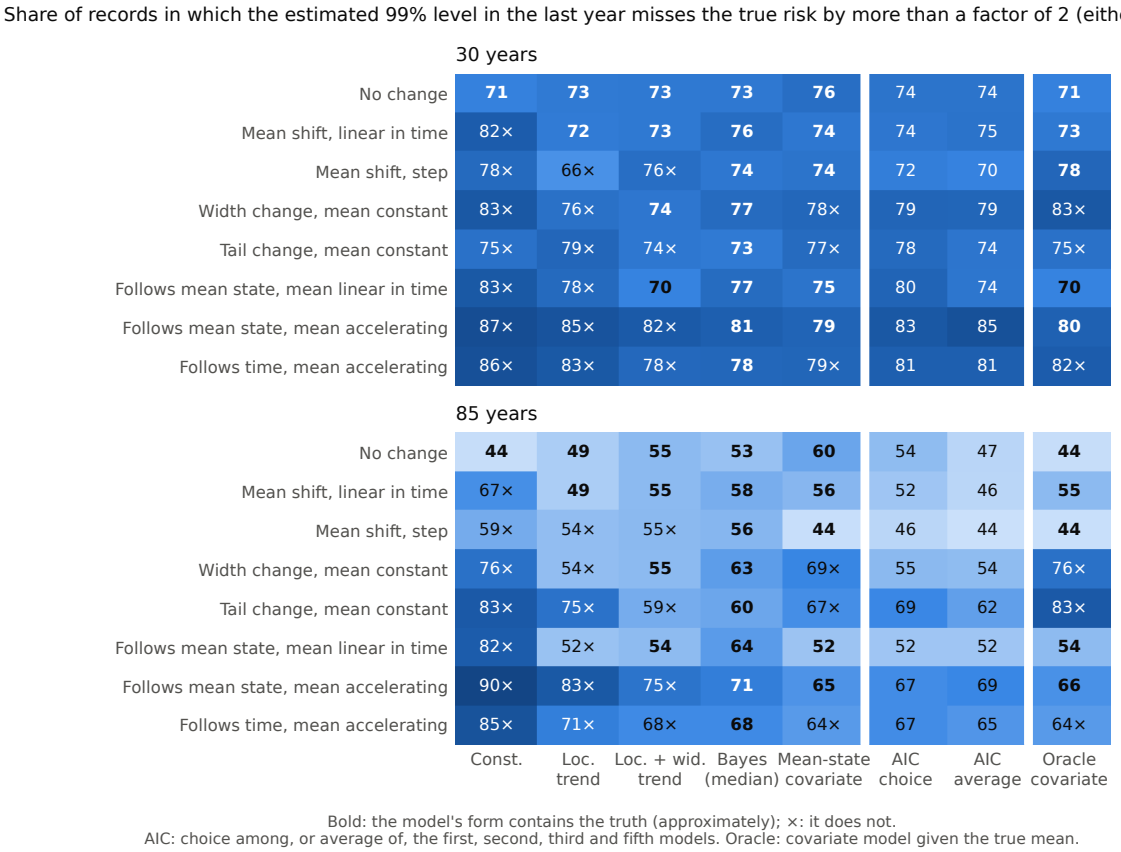


Figure 14: Share of records in which the estimated 99% level in the last year misses the true risk by more than a factor of 2 in either direction, by true process (rows) and fitted model (columns). Bold: the model's form contains the true process (approximately); ×: it does not.

	Missed by > 2× either way, % of records							AIC picks, % of records			
True process	Const.	Loc.	Loc.+wid.	Bayes	Cov.	AIC	AIC avg.	Const.	Loc.	Loc.+wid.	Cov.
30 yr											
No change	**71**	**73**	**73**	**73**	**76**	74	74	64	16	5	16
Mean shift, linear in time	82	**72**	**73**	**76**	**74**	74	75	40	44	8	9
Mean shift, step	78	66	76	**74**	**74**	72	70	27	53	5	15
Width change, mean constant	83	76	**74**	**77**	78	79	79	29	40	15	16
Tail change, mean constant	75	79	74	**73**	77	78	74	29	36	15	20
Follows mean state, mean linear in time	83	78	**70**	**77**	**75**	80	74	12	60	11	17
Follows mean state, mean accelerating	87	85	82	**81**	**79**	83	85	18	38	10	34
Follows time, mean accelerating	86	83	78	**78**	79	81	81	13	48	12	27
85 yr											
No change	**44**	**49**	**55**	**53**	**60**	54	47	69	11	4	17
Mean shift, linear in time	67	**49**	**55**	**58**	**56**	52	46	22	59	2	17
Mean shift, step	59	54	55	**56**	**44**	46	44	6	20	2	72
Width change, mean constant	76	54	**55**	**63**	69	55	54	6	71	17	6
Tail change, mean constant	83	75	59	**60**	67	69	62	6	38	38	18
Follows mean state, mean linear in time	82	52	**54**	**64**	**52**	52	52	2	66	5	26
Follows mean state, mean accelerating	90	83	75	**71**	**65**	67	69	3	27	4	66
Follows time, mean accelerating	85	71	68	**68**	64	67	65	2	48	8	42

Table 3: Using the wrong model. Left: share of records (%) in which the estimated 99% level in the last year misses the true risk by more than a factor of 2 either way; bold where the model's form contains the true process. Right: how often AIC selects each candidate. Loc.: location trend; Loc.+wid.: location and width trend; Cov.: mean-state covariate. Appendix Table 5 splits the misses into under- and overstatement.

Findings.

- **The wrong driving variable can cost 10–20 percentage points at 85 years.** When the distribution follows an accelerating mean state, the location + width trend misses 75% of records and the covariate model 65%. When the width changes while the mean stays constant, the covariate model, whose covariate then carries no signal, misses 69% and the location + width trend 55%.
- **Unneeded flexibility costs about as much.** With no change, the covariate model misses 60% of records against 44% for the constant fit, because it fits a relation to the year-to-year noise of the annual mean; the location + width trend misses 55%.
- **Where the two kinds of model coincide, the choice hardly matters**: with the mean linear in time, with or without the distribution following it, the location trend, location + width trend and covariate model miss 49–56% of records.
- **A step favours the covariate model** (44% against 54–55% for the time trends), because the smoothed annual mean follows the step and a straight line cannot.
- **For 30-year records the choice matters little**: every model misses 66–87% of records, and within each true process the models differ by 5–13 points, while sampling noise alone causes 71%.
- **When the mean accelerates, the mean-state model loses little even if the width follows time.** In the time-driven counterparts of the accelerating warming cases the covariate model's assumption about the width is wrong, yet it misses 64% of 85-year records, against 68% for the location + width trend and the Bayesian model and 71% for the location trend.

The time-trend models do better than when the width follows the mean (68% against 75% for the location + width trend), and the covariate model about the same (64% against 65%). Following the accelerating mean, which a straight line in time cannot, matters more than the driver of the width, which differs little between the two (Figure 15). The exception is the heavy-tailed t parent, whose change is mostly in the tail: there the location + width trend does better (48% against 54%). Our parametric time models are linear; a time model with a curved mean might do better, but we have not tested one.

Can the data tell? AIC selects among the constant, location-trend, location + width trend and covariate models (Table 3, right). With 85 years it picks the covariate model in 66% of the accelerating mean-state cases and 72% of the step cases, but also in 17% of records with no change. It almost never detects a width change: when only the width changes it picks the location trend in 71% of records and the location + width trend in 17%. When the width follows time rather than the mean, it picks the covariate model less often (42% against 66%), so it leans the right way, but it cannot reliably tell the two apart. With 30 years it picks the constant fit or the location trend in most records whatever the truth.

Picking the lowest-AIC model is never far off, but it is sometimes 10 points worse than the best single model: with no change it misses 54% of 85-year records against 44% for the constant fit, and after a tail change 69% against 59%. Averaging the four models with Akaike weights does better (Table 3). At 85 years it is within 4 points of the best single model for every true process, and when the mean moves linearly it beats all of them (46% against 49% for the best). By contrast, every single model, including the Bayesian one, is 11–30 points worse than the best for at least one process. On the one-sided score of Section 4 the average matches the location trend over the standard cases (30% understated at 85 years, Table 2), and in the covariate-model cases it lies between the location + width trend and the covariate model (52% against 60% and 45%). With 30 and 45 years the differences between all the choices are small, and the average is within 6 points of the best model throughout.

What to do when we do not know.

- **Do not expect the record to reveal the driver.** The time and mean-state models coincide whenever the mean moves linearly in time, and AIC rarely detects a width change. Whether the change follows time or the mean state has to come from physical reasoning or from evidence outside the record.
- **Average rather than choose.** An AIC-weighted average of the constant, time-trend and mean-state models gives up a few points against the best model for the true process, whatever that is, and avoids the large losses of committing to a wrong one.
- **When the mean clearly moves nonlinearly, include a mean-state model.** With an accelerating mean or a step, the covariate model did best whether or not the width followed the mean.
- **Report the spread between plausible models** as part of the uncertainty of the current return level, alongside the sampling uncertainty of each.
- **With 30 years, the model matters less than the data.** Two-thirds or more of 30-year records miss the current 99% level by a factor of 2 whatever is fitted; only longer records or

additional information (other sites, or more declustered values per year) are likely to change that.

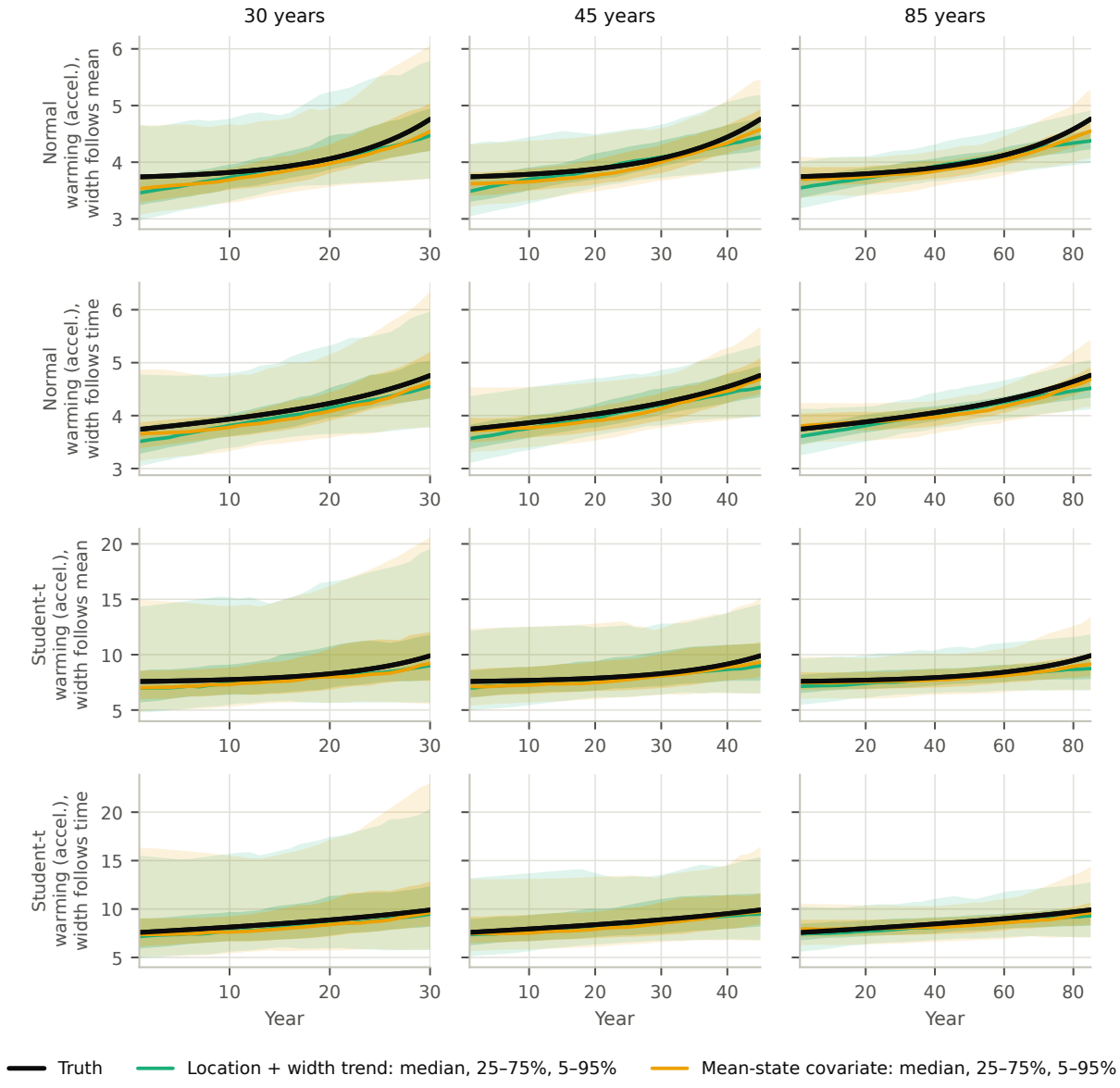


Figure 15: The same accelerating warming with the width (and for the t the tail) following the mean state ("width follows mean") or changing linearly in time ("width follows time"); the first- and last-year distributions are the same in each pair. Location + width trend (aqua) against the mean-state covariate model (yellow), which have the same number of parameters. Median (line), 25–75% and 5–95% of the estimated 99% level over 100 records, truth in black.

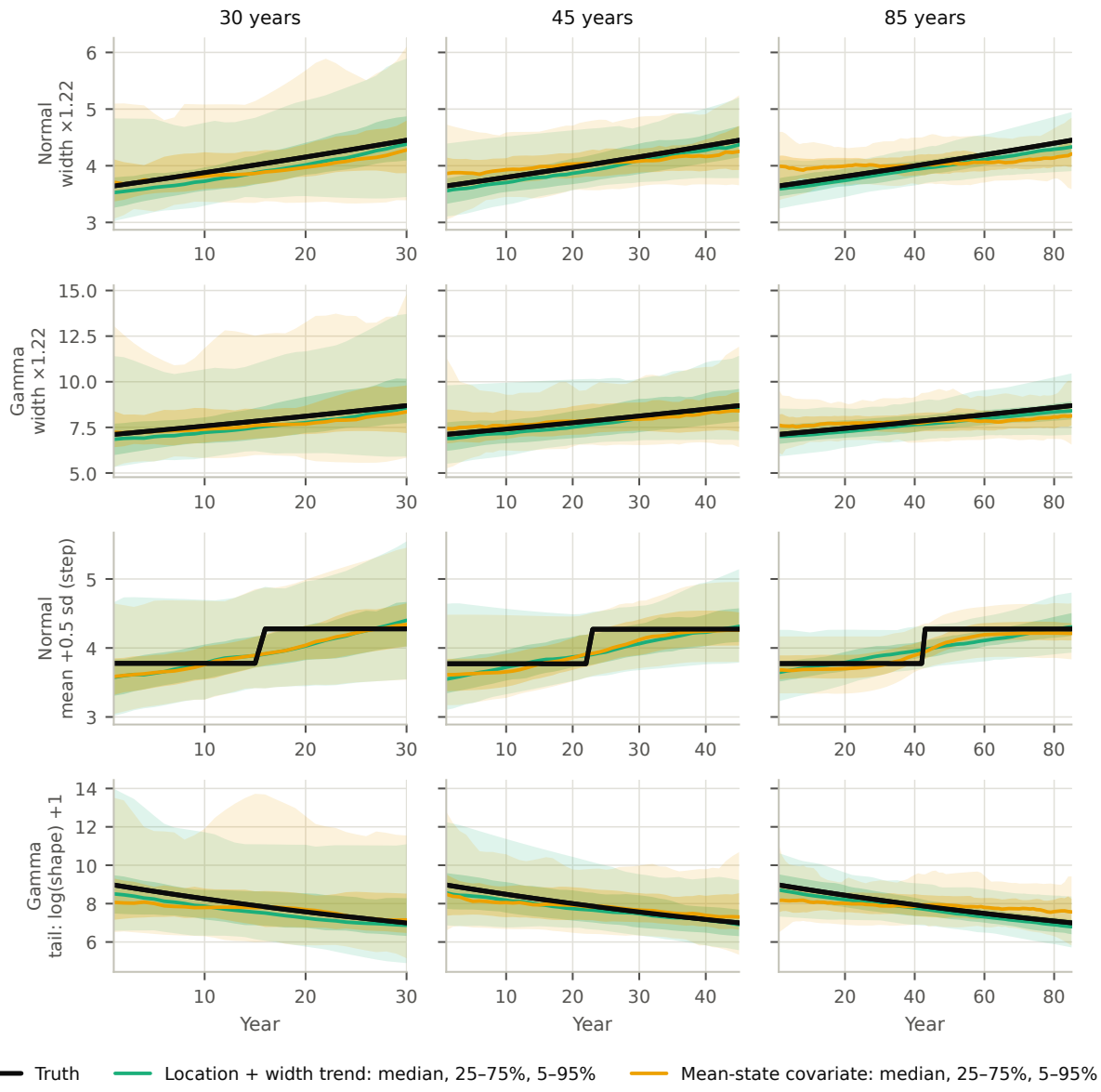


Figure 16: Cases in which the mean state does not track the change. For a width or tail change at constant mean the covariate model (yellow) has no signal and stays nearly flat; for a step it follows the smoothed step. Location + width trend in aqua. Median, 25–75% and 5–95% over 100 records, truth in black.

The covariate-model cases in detail. In these cases the width and tail of the daily distribution change with the mean, and the mean itself rises along an accelerating path. On the one-sided score of Section 4, at 85 years the location trend understates the current risk twofold in 71% of records, the location + width trend in 60% and the Bayesian model in 57%; the covariate model does markedly better (45%), close to its oracle version with the true mean path (43%). Two things contribute: the covariate captures the acceleration, which a straight line in time cannot, and it is estimated from all daily values, not only the extremes, so it is much less noisy than the extremes themselves. Letting the tail index also depend on the covariate makes things worse (57%), because

the tail index then follows the noise in the covariate.

Figures 17 and 18 show the paths. The covariate model (yellow) follows the accelerating rise; the location trend (blue), a straight line, cuts across it and ends well below the truth (by 0.4 daily sd for the normal parent, 40% of the true change), and this shortfall does not shrink with record length, because it comes from the form of the model, not from noise. The covariate model's median also ends somewhat low (by 0.2), but so does its oracle version with the true mean as covariate (by 0.14): this is mostly the downward tendency of a typical high-quantile estimate seen already in the no-change case. Its bands widen at the end of short records, where the smoothed covariate is least certain. For the t parent, whose tail gets heavier as it warms, the shortfall is larger (0.8 against 1.5 for the location trend), because the model holds the tail index constant.

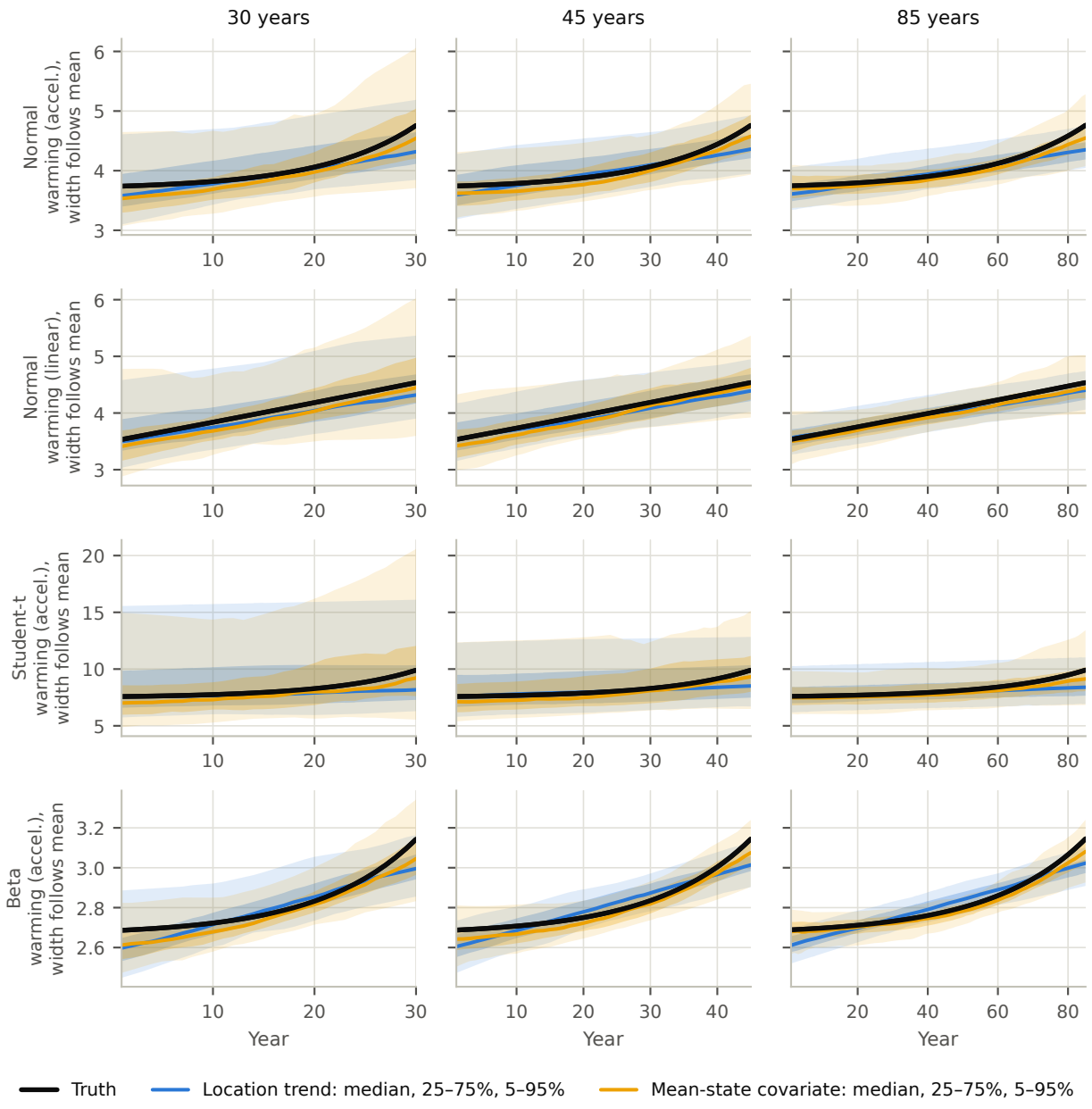


Figure 17: Temperature-like covariate-model cases: the mean rises, and the width (and for the t and beta the tail) follow it. Location trend (blue) against the mean-state covariate model (yellow). Median (line), 25–75% (dark band) and 5–95% (light band) of the estimated 99% level in each year over 100 records, with the truth in black. Columns: record length; the total change over the record is the same in each column. Units: baseline daily sd.

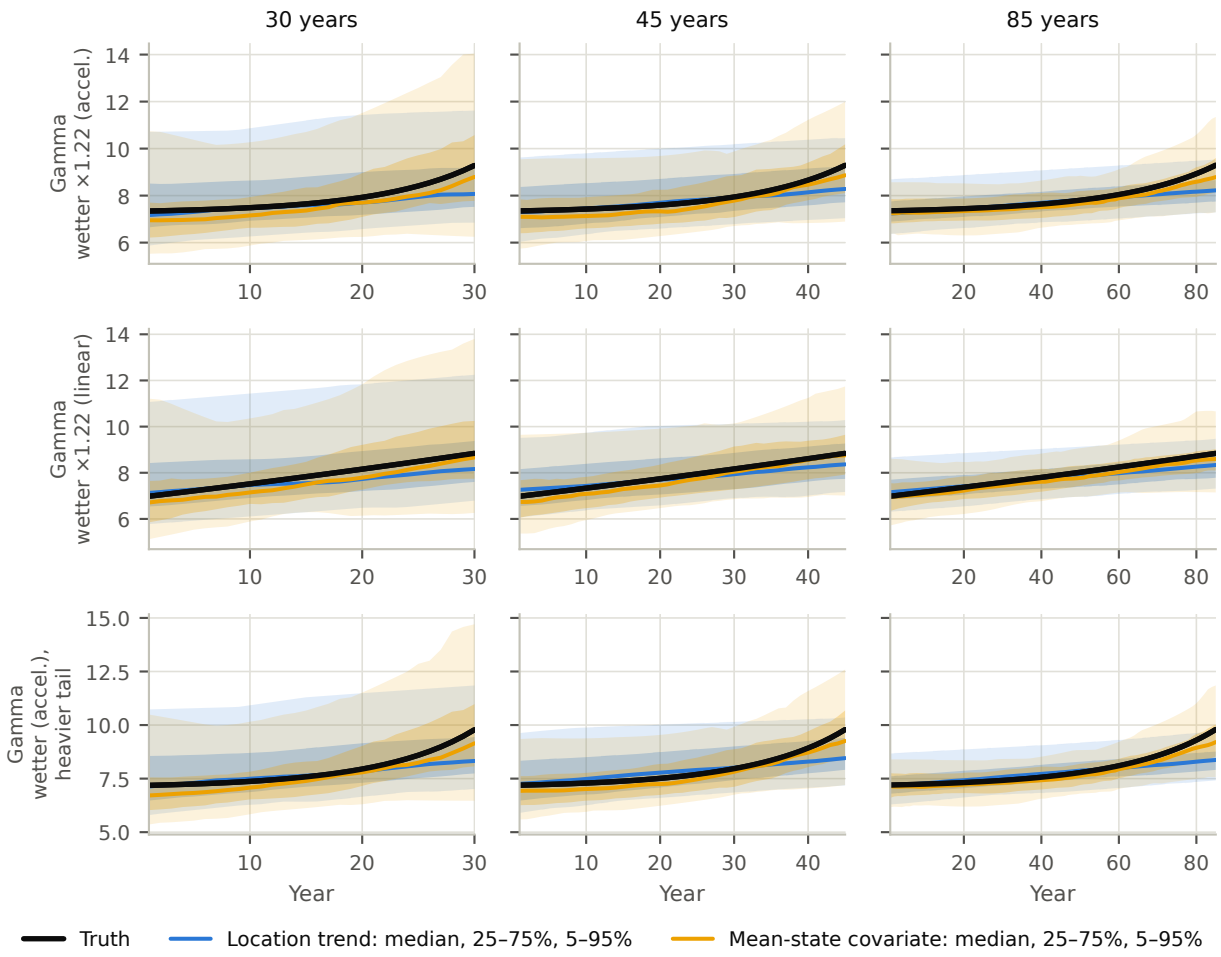


Figure 18: Precipitation-like covariate-model cases (gamma): every quantile scales with the mean, and in the last row the tail also gets heavier. Location trend (blue) against the mean-state covariate model (yellow). Median (line), 25–75% (dark band) and 5–95% (light band) of the estimated 99% level in each year over 100 records, with the truth in black. Columns: record length; the total change over the record is the same in each column. Units: baseline daily sd.

6 The Bayesian random-walk model

Credible intervals. The 90% intervals for the last year's $X_{0.99}$ contain the truth in 94–100% of records, so they are conservative (Figure 19). Their median width is 1.3 to 3 times the 5–95% range of the actual errors of the posterior median. The intervals are honest but too wide to be informative for short records.

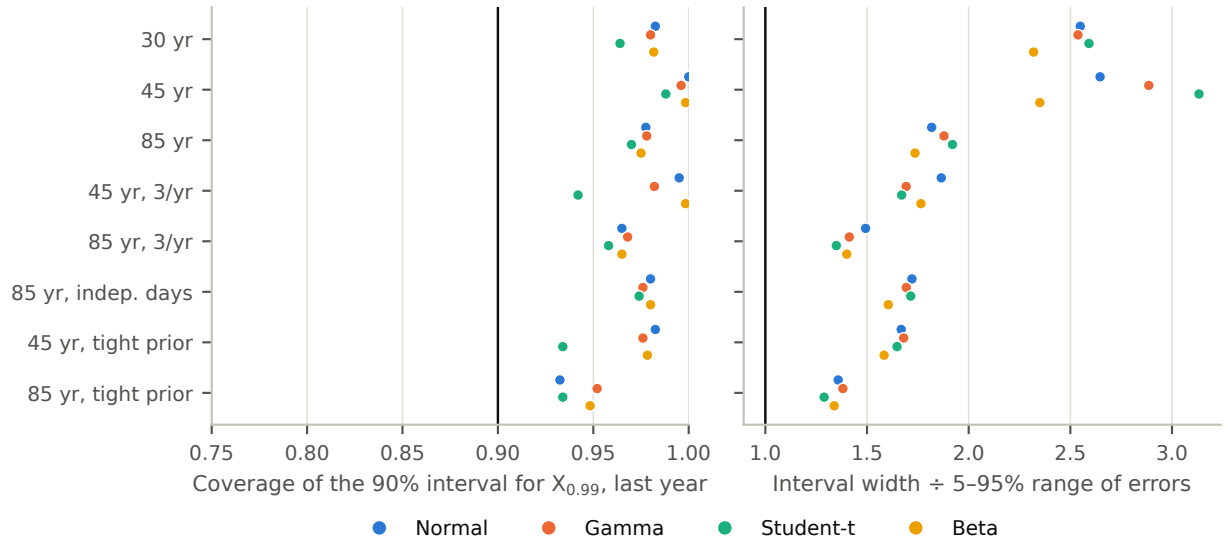


Figure 19: Bayesian model. Left: coverage of the nominal 90% interval for the last year's 99% level. Right: median interval width relative to the 5–95% range of the actual errors (1 = right width). One dot per parent; standard cases.

Convergence. Under the authors' prior most fits have some divergent transitions (median 9 of 4,000 at 85 years and 81 at 30 years). With one value per year, R-hat exceeds 1.05 in fewer than 1% of fits. With three values per year it exceeds 1.05 in 10–12% of fits, almost always far above (R-hat > 1.5, chains in different modes), and these are concentrated in the bounded beta parent (25–32% of its fits). Our chains are shorter than the authors', but disagreement between chains of this size is not a matter of chain length. The unconverged fits overstate the risk more often than the converged ones (37–49% against 19–32% overstated twofold), but they are a minority, and excluding them changes the pooled shares by at most 2 points.

Prior sensitivity. Shrinking the random-walk scales for ξ, α and s to 0.25 nearly eliminates the convergence problems (R-hat ≤ 1.05 in every fit, median 1–6 divergences). It improves accuracy slightly at 85 years (35% against 39% understated) and leaves 45 years unchanged (34%), and it brings interval coverage closer to nominal (93–98%). See Appendix Figure 22. Figure 20 shows the cost: the tighter prior narrows the bands, but it also makes the paths stiffer, so for the t tail change the median follows less of the early, high values. Figure 21 compares both priors with the covariate model in the covariate-model cases. There the tighter prior is slightly worse (60% understated at 85 years against 57%, and 45% for the covariate model), and its intervals cover the truth in only 88% of records (80–92% by case), because its stiffer paths lag the acceleration at the end of the record.

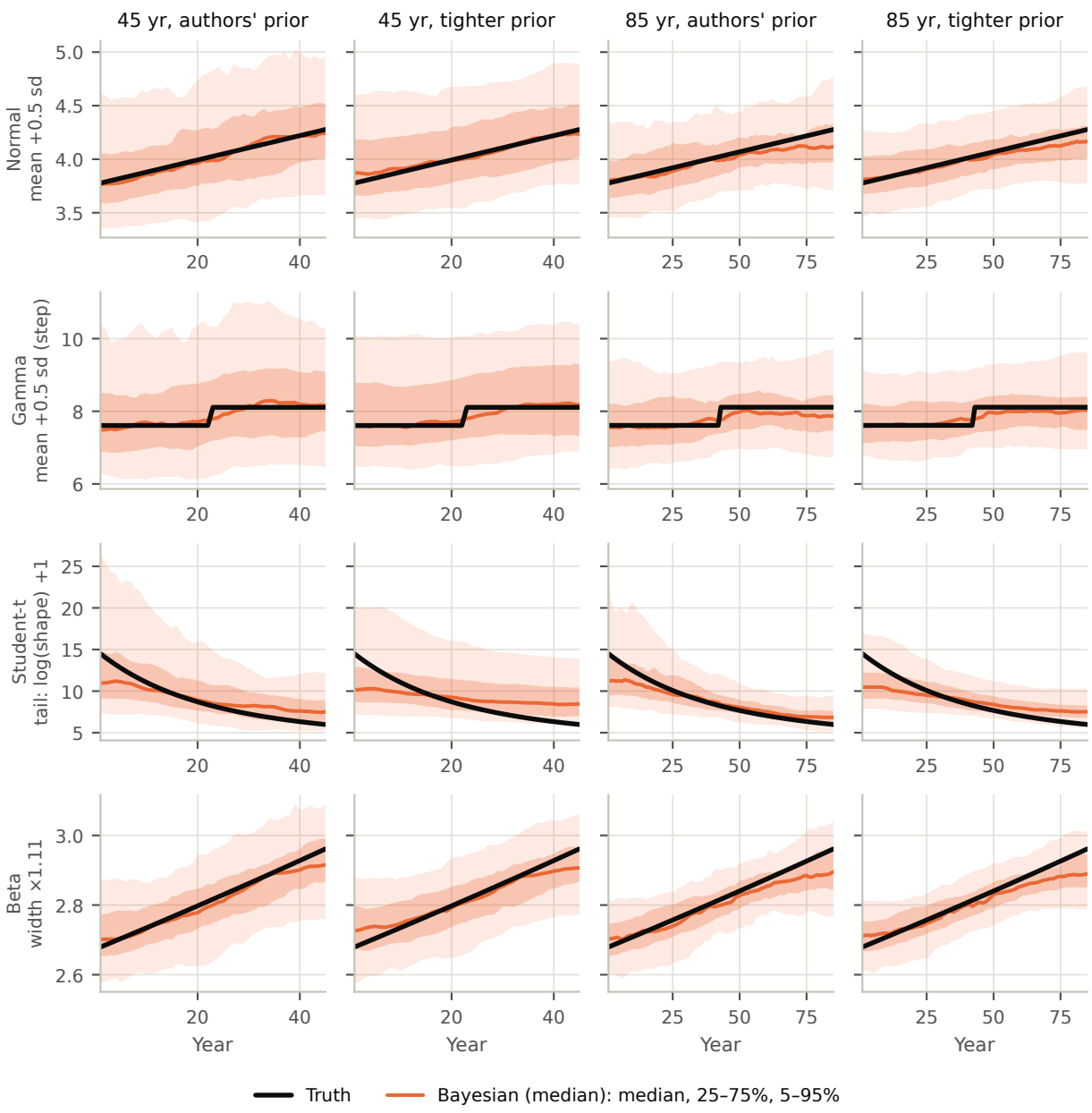


Figure 20: Bayesian posterior median with the authors' prior and a tighter prior on the random-walk scales, at 45 and 85 years. Median (line), 25–75% (dark band) and 5–95% (light band) of the estimated 99% level in each year over 100 records, with the truth in black. Units: baseline daily sd.

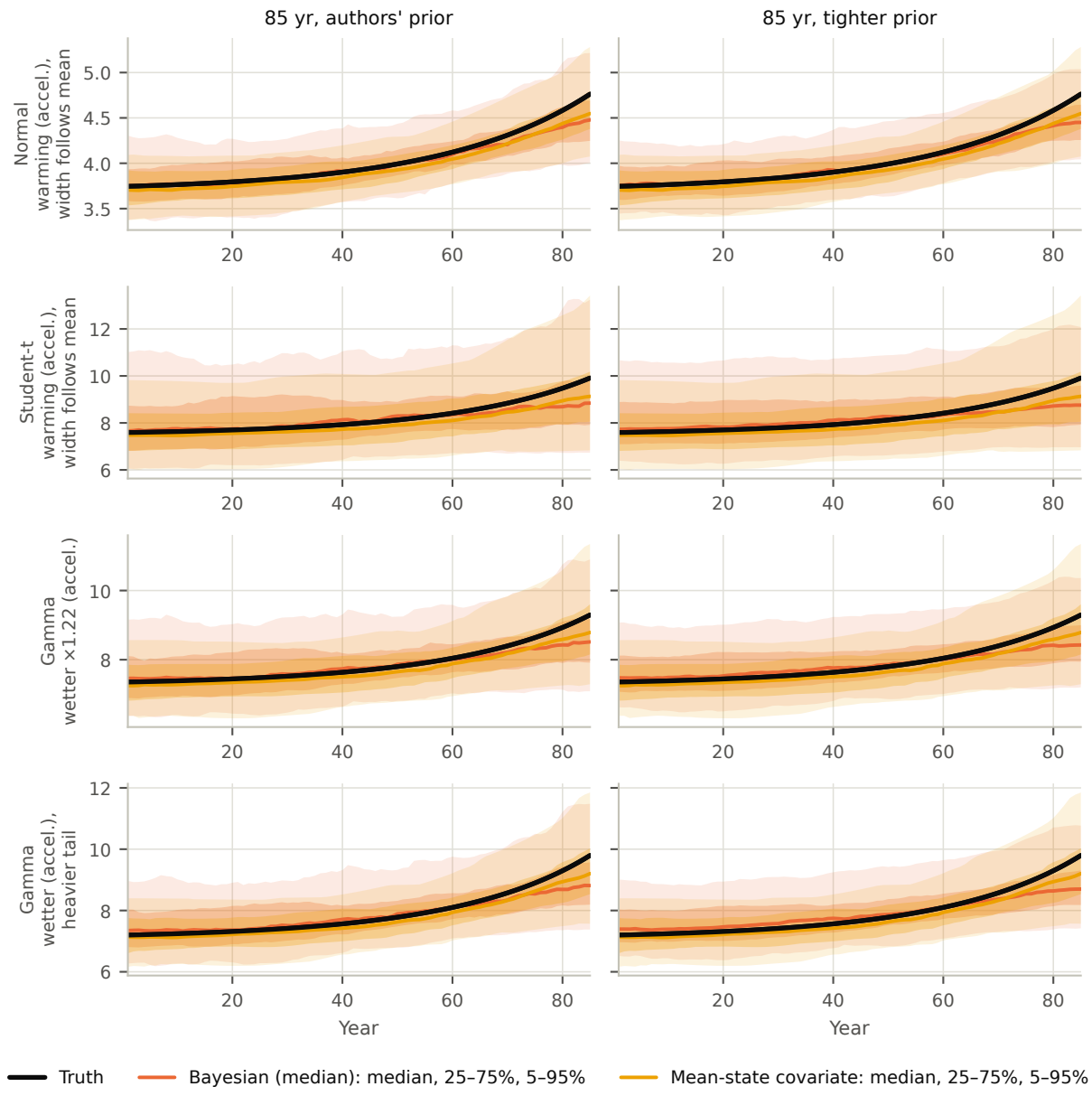


Figure 21: Covariate-model cases at 85 years: Bayesian posterior median (orange) with the authors' and the tighter prior, and the mean-state covariate model (yellow). Median (line), 25–75% (dark band) and 5–95% (light band) of the estimated 99% level in each year over 100 records, with the truth in black. Units: baseline daily sd.

Time reversal. The model's prior is symmetric in time, so fitting a record in reverse order should give the reversed posterior paths. For four test cases the largest difference between the forward and reversed posterior medians of $X_{0.99}$ was 0.003–0.11, at most 1% of the width of the 90% interval, consistent with MCMC noise.

7 Limitations

- The synthetic data have no seasonal cycle, a single AR(1) dependence structure, and changes that are smooth or a single step. Real daily data have seasonality, longer memory and clustering of extremes, which would mostly make things harder.
- The covariate-model cases are, by construction, cases where that model is right, and the covariate is the local mean of the same series. With an external covariate (e.g. global mean temperature) the gain would depend on how closely the local extremes follow it.
- 100 records per case gives standard errors of 3–4 percentage points on the pooled shares; differences between estimators of less than about 5 points should not be over-interpreted.
- Our MCMC runs are shorter than the authors' ($4 \times 1{,}000$ against $4 \times 10{,}000$ draws). The accuracy of point estimates and interval coverage should be robust to this; the convergence failures with three values per year are not a matter of chain length.
- We have not yet run the paper's joint L^* test, the exceedance (threshold) versions of the tests, or the Bayes factors that compare "more events" with "a shifted distribution".

8 Next steps

1. Apply the tests and the location-trend, covariate and Bayesian estimators to ERA5 (85 years) at selected grid cells or regions, with declustered values, and compare with the synthetic-data error rates reported here.
2. Add seasonality to the synthetic data, and an external covariate (global mean temperature) to the covariate model.
3. Report uncertainty for the current return level from the MLE estimators (profile likelihood or bootstrap) and check its coverage as we did for the Bayesian intervals.

A Cases and supplementary results

Set	Parent	Change over the record	$\Delta X_{0.99}$
Standard	Normal	no change	+0.00
	Normal	mean +0.5 sd (linear)	+0.50
	Normal	mean +0.5 sd (step)	+0.50
	Normal	width ×1.22 (linear)	+0.81
	Gamma	no change	+0.00
	Gamma	mean +0.5 sd (linear)	+0.50
	Gamma	mean +0.5 sd (step)	+0.50
	Gamma	width ×1.22 (linear)	+1.58
	Gamma	tail: log(shape) +1 (linear)	-1.97
	Student-t	no change	+0.00
	Student-t	mean +0.5 sd (linear)	+0.50
	Student-t	mean +0.5 sd (step)	+0.50
	Student-t	width ×1.22 (linear)	+1.64
	Student-t	tail: log(shape) +1 (linear)	-8.51
	Beta	no change	+0.00
	Beta	mean +0.25 sd (linear)	+0.25
	Beta	mean +0.25 sd (step)	+0.25
	Beta	width ×1.11 (linear)	+0.28
	Beta	tail: log(shape) +0.2 (linear)	-0.11
	Beta	position: log α +0.5 (linear)	+0.07
Covariate model	Normal	warming +0.5 sd, width & tail follow (accel.)	+1.02
	Normal	warming +0.5 sd, width & tail follow (linear)	+1.00
	Gamma	wetter: all quantiles ×1.22 (accel.)	+1.94
	Gamma	wetter: all quantiles ×1.22 (linear)	+1.86
	Gamma	wetter ×1.22, heavier tail (accel.)	+2.60
	Student-t	warming +0.5 sd, width & tail follow (accel.)	+2.32
	Beta	warming +0.25 sd, width & tail follow (accel.)	+0.46
Time-driven	Normal	warming +0.5 sd, width & tail linear in time (accel.)	+1.02
	Student-t	warming +0.5 sd, width & tail linear in time (accel.)	+2.32
	Beta	warming +0.25 sd, width & tail linear in time (accel.)	+0.46

Table 4: Stage C cases. $\Delta X_{0.99}$: true change in the 99% level over an 85-year record (baseline daily sd). "log(shape)" is the log of the gamma shape a, the t degrees of freedom ν, or the beta β; increasing it lightens the tail of the gamma and t.

True process	Const.	Loc.	Loc.+wid.	Bayes	Cov.	AIC	AIC avg.
Understated > 2×, 85 years							
No change	**27**	**32**	**36**	**36**	**38**	32	29
Mean shift, linear in time	61	**32**	**36**	**44**	**38**	36	31
Mean shift, step	52	16	19	**37**	**25**	26	23
Width change, mean constant	73	47	**37**	**50**	54	41	40
Tail change, mean constant	0	25	34	**21**	20	36	27
Follows mean state, mean linear in time	80	44	**36**	**50**	**35**	38	36
Follows mean state, mean accelerating	89	81	69	**60**	**49**	55	58
Follows time, mean accelerating	84	67	59	**55**	42	54	53
Overstated > 2×, 85 years							
No change	**18**	**17**	**19**	**17**	**23**	22	18
Mean shift, linear in time	6	**17**	**19**	**14**	**18**	16	15
Mean shift, step	6	38	36	**19**	**19**	21	21
Width change, mean constant	2	7	**18**	**12**	15	14	14
Tail change, mean constant	82	50	25	**39**	47	32	35
Follows mean state, mean linear in time	2	8	**19**	**13**	**17**	14	16
Follows mean state, mean accelerating	1	2	5	**12**	**16**	12	11
Follows time, mean accelerating	1	4	9	**13**	22	14	12

Table 5: The misses of Table 3 at 85 years, split into records with the risk understated (true exceedance > 2%) and overstated (< 0.5%) at least twofold. Bold: the model's form contains the true process.

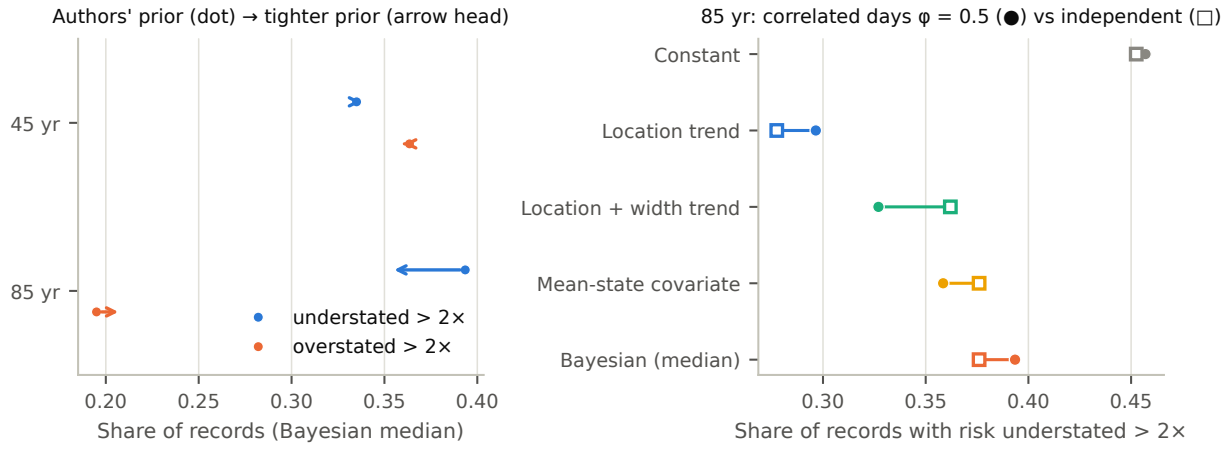


Figure 22: Left: Bayesian model with the authors' prior and with a tighter prior on the random-walk scales. Right: shares of 85-year records with the risk understated twofold with correlated ($\phi = 0.5$) and independent daily values.

References

Carpenter, B., et al. (2017). Stan: a probabilistic programming language. *Journal of Statistical Software* 76(1).

Coles, S. (2001). *An Introduction to Statistical Modeling of Extreme Values*. Springer.

Müller, U. K. and M. W. Watson (2024). Time varying extremes. *Review of Economics and Statistics*, doi:10.1162/rest_a_01528.

Nyblom, J. (1989). Testing for the constancy of parameters over time. *Journal of the American Statistical Association* 84, 223–230.

Philip, S., et al. (2020). A protocol for probabilistic extreme event attribution analyses. *Advances in Statistical Climatology, Meteorology and Oceanography* 6, 177–203.